
Software Requirements Specification

for

SoftwareMon

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version
Everybody	3 Feb 2021 - 14 Feb 2021	Functional Requirements Non-Functional Requirements Use Case Diagrams and Descriptions Data Dictionary Initial UI Mockups	1.0
Everybody	15 Feb 2021 - 1 Mar 2021	Refined Use Case Diagrams and Descriptions Class Diagram Sequence Diagrams Dialog Map	2.0
Everybody	29 Mar 2021	Refined Use Case Diagrams and Descriptions Refined Class Diagram Refined Sequence Diagrams Refined Dialog Map System Architecture Application Skeleton	3.0
Everybody	31 Mar 2021	Refined Use Case Diagrams and Descriptions Refined Class Diagram Refined Sequence Diagrams Refined Dialog Map Black Box Testing White Box Testing Demo Script	4.0
Everybody	1 Apr 2021 - 13 Apr 2021	Introduction of SRS External Interface Requirements Design Pattern Conclusion Refined Functional Requirements Refined Class Diagram Refined Sequence Diagram Refined Dialog Map Refined Non-Functional Requirements Refined Use Case Diagrams and Descriptions Refined System Architecture Design Refined Data Dictionary	5.0

1. Introduction

1.1 Purpose

This document specifies the software requirements of SoftwareMon version 1.0, a Unity game application. This document will cover the main functionalities of the application, the various user interfaces, as well as the constraints.

1.2 Document Conventions

In this SRS, a different font or style does not necessarily indicate the importance or special significance of that particular sub-chapter. Such styles serve only to ensure proper formatting within the document. The priority of each specific requirement is written and elaborated in words in their own respective sections.

1.3 Intended Audience and Reading Suggestions

This document is intended for developers, project managers, stakeholders, and users, who will be using this documentation to further understand the SoftwareMon game application. The above-mentioned user groups are encouraged to read the documentation in sequential order and use the appendix given to fully understand the architecture of the PAS application. Developers are required to refer to the system features and requirements in order to develop the application in line with the intended specifications. This group is advised to refer to “Overall Description” to the “Other Requirements” section of the proposal. The quality assurance team will need to go look at the functional requirement to generate the use case scenarios. This group is advised to refer to the “System Features” section of the proposal.

1.4 Product Scope

SoftwareMon is a Unity game application that allows students to experience learning through an entertaining, interactive and competitive medium by exposing them to a variety of software engineering modules to enhance their technical knowledge and concepts. SoftwareMon also enables teachers to track the progress and capabilities of their students through in-game statistics.

1.5 References

- [1] “ISO/IEC/IEEE International Standard – Systems and software engineering – Life cycle processes – Requirements engineering,” ISO/IEC/IEEE 29148:2011€, pp. 1–94, Dec. 2011.
- [2] Javadoc tool. (n.d.). Retrieved April 1, 2021, from <http://www.oracle.com/technetwork/java/javase/documentation/index-jsp-135444.html>

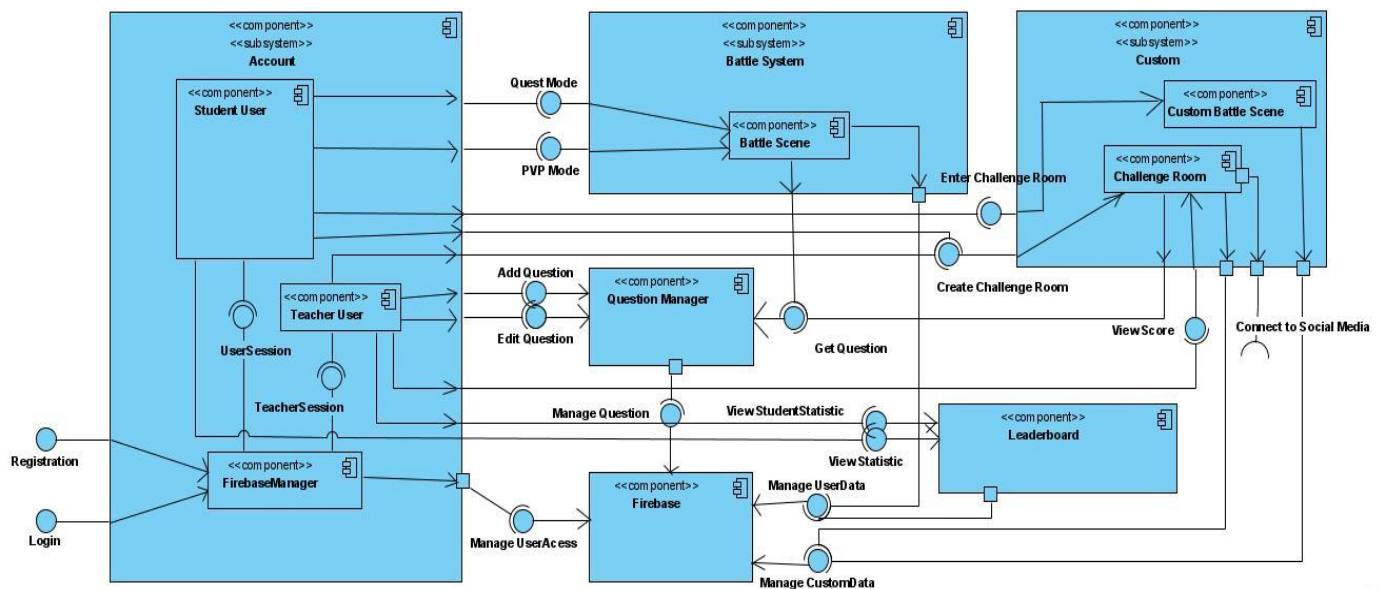
2. Overall Description

2.1 Product Perspective

Software engineering modules are often mundane and challenging for both students and teachers alike. For students, an un-engaging curriculum hinders their learning and limits their potential. For teachers, limited assessments and teacher-student interactions make it challenging for them to properly assess the learning progress of their students. With these challenges in mind, our team has come up with an application to make learning fun for everyone, students and teachers alike.

Our application is a game where students practise questions from different software engineering modules taught in school. These questions subsequently increase in difficulty as their mastery of the module increases. Our game allows students to engage in learning through an immersive in-game experience. Students explore different worlds (each representing a software engineering module) in the game's quest mode, face off against NPC trainers and even compete against their peers in an offline Player-vs-Player (PVP) battle system in this fantasy game. The stars and points accumulated in both game modes will be stored in a real-time database, in which users can view their statistics in an interface within the game. Furthermore, the game provides a link for users to access the discussion forum which we have set-up and share their enquiries or announcements to others, providing an interactive experience for all.

The following diagram illustrates the major components of the overall system, subsystem interconnections, and external interfaces:

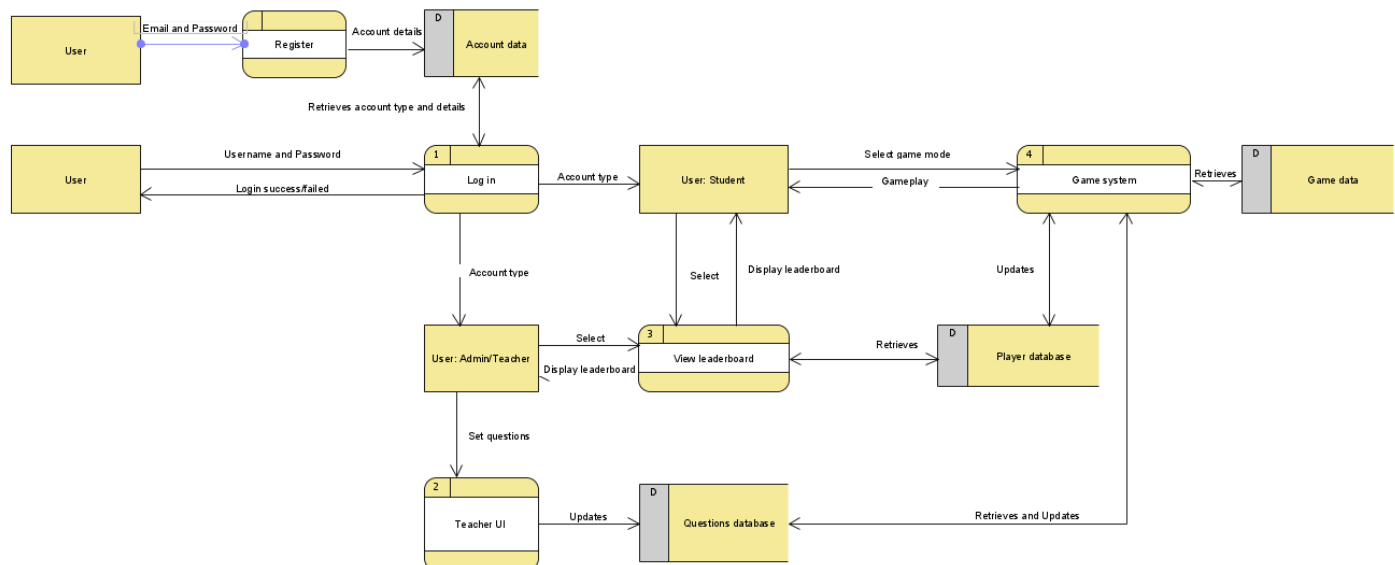


Powered By/Visual Paradigm Community Edition

2.2 Product Functions

1. The application must have an interface for new users to register a new user account.

2. The application must have an interface for users to login.
3. The application shall have a role-playing game theme.
4. The application shall have 4 characters for users to choose from.
5. The application shall have a quest mode that consists of 3 worlds for users to learn different phases of the life cycle of Software Engineering.
6. The application shall have a PVP mode for users to challenge each other offline.
7. The application shall have a custom mode for users to challenge each other with their own set of selected questions.
8. The application must have a leaderboard interface to display the PVP scores of the users.
9. The application must have an interface for teacher users to create and issue assignments to student users.
10. The application must have an interface for teacher users to view the scores of student users who have attempted the assignment..
11. The application shall provide a discussion forum for teacher and student users to participate in discussions.
12. The application shall have an interface for teacher users to track the statistics of student users.



2.3 User Classes and Characteristics

While SoftWareMon accommodates all levels of users who show interest in learning software engineering modules through gaming, the focus of our application caters to two user groups as follows.

User Class	Description
Students	- Given that our social game is requested by the Teaching, Learning and Pedagogy Division (TLPD) of NTU with the intention of gamifying the teaching and learning of software engineering courses, students are the focus of our application. - These students are likely to be the most active on our application to learn about software engineering courses in an interactive and engaging way. Therefore, they are more likely to be found on the Custom mode, PVP mode or story mode. - At the same time, teacher-related functions are restricted for students. For example, they will not be able to access the statistics page showing the personal statistics of each student.
Teachers	- Teachers are likely to be logging into the social game on a periodic but consistent basis to monitor the progress of students, or to distribute assignments. - Hence, they are more likely to be custom game modes to create assignments or the statistics page to track the individual progress of their students, and less likely to be on the story mode of PVP mode of the social game.

2.4 Operating Environment

Operating Platform:

- Development Platform: Unity Game Engine
- Operating System Platform: Windows, Mac OS
- Software Version: The game shall be compatible with any OS above Windows 7 and Mac OS X 10.10

Backend:

- Data is stored on Firebase Realtime Database
- Authentication is conducted through Firebase Authenticator

2.5 Design and Implementation Constraints

Since this application requires users to log in, the company would need to maintain a secure database which allows for the passwords of their users to be safely secured. Furthermore, since the user is required to key in their own personal information, the Personal Data Protection Act (PDPA) will need to be strictly adhered to, to ensure that the particulars of users are safely secured and accessed only when necessary.

2.6 Assumptions and Dependencies

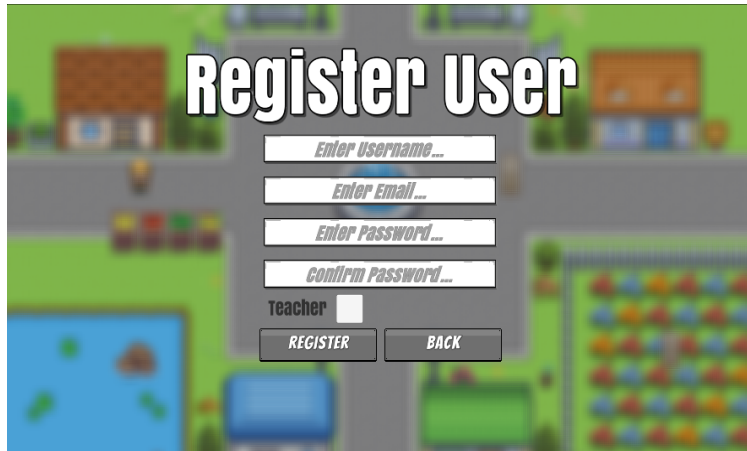
It is assumed that users of the social game are always connected to the Internet and have a stable internet connection.

The question banks stored in the database are only representative of the current syllabus of NTU software engineering modules. Therefore, the database will have to be updated should there be any changes in the school syllabus.

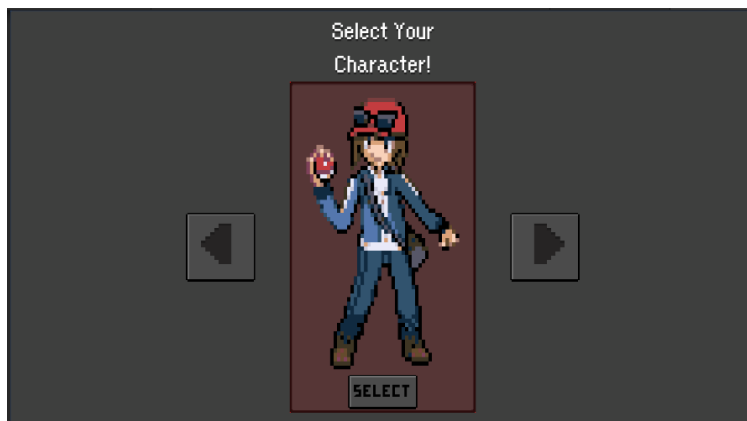
3. External Interface Requirements

3.1 User Interfaces

Display	Page Summary
	Login page • This is the login page for existing/registered users to login. • The user can input his/her email address and password before clicking on the “Login” button.

**Register page**

- This is the registration page for new users to register their user account. The design of the registration page allows users to input the following details:
 - Username
 - Email Address
 - Password
 - Password Confirmation
 - Teacher/Not Teacher
- Upon entering the details, users can complete the registration process by clicking on the “Register” button, or return to the login page by clicking on the “Back” button.

**Character Selection page**

- This is the character selection page, which displays 4 avatars for the user to choose from.
- The selected avatar will be reflected in the subsequent sections of the game.



Game Mode page

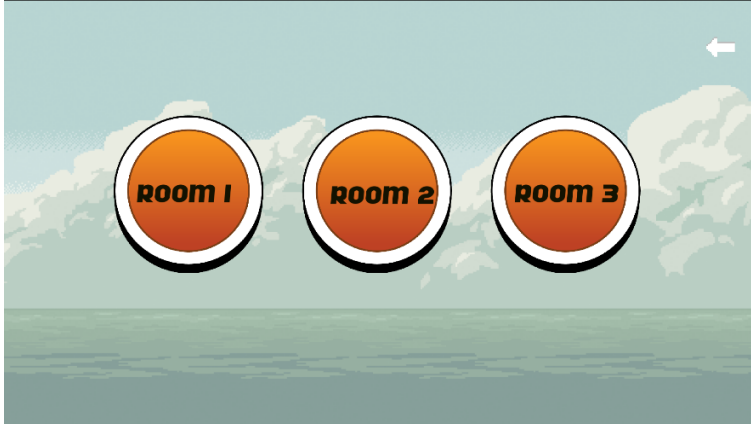
- This is the game mode page, which allows the user to access 4 sections of the game:
 - Quest
 - PvP
 - Custom
 - View Profile
- The Quest mode allows users to learn software engineering modules in a fun and engaging way.
- The PvP mode allows users to socialise with other users and compete in their knowledge of software engineering concepts.
- The custom mode allows users to create their own challenges and share them with others.



View Profile page

- This is the profile page after clicking View Profile.
- The profile page displays the following statistics of the user:
 - Total Stars
 - Total Points
- The profile page has the following filters to allow the user to filter his/her statistics:
 - World

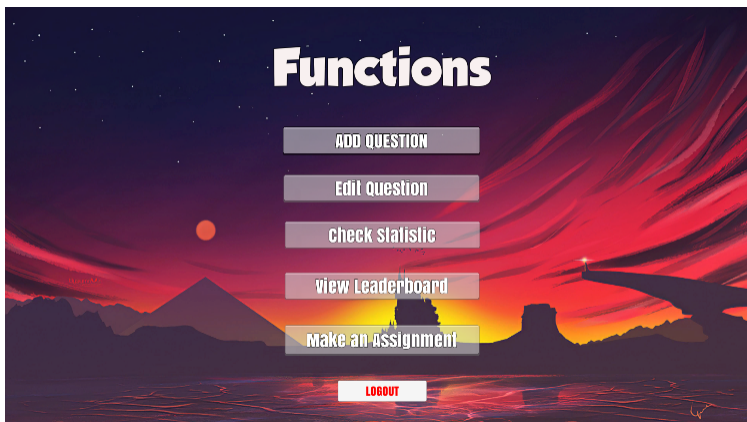
	○ Section ● The main menu page allows the user to view leaderboard or go back to the game mode game.																																
 <table><thead><tr><th>Username</th><th>Rank</th><th>Worldsection</th><th>Points</th></tr></thead><tbody><tr><td>shaun</td><td>1</td><td>All</td><td>507</td></tr><tr><td>freeacc0002</td><td>2</td><td>All</td><td>462</td></tr><tr><td>Aangtara</td><td>3</td><td>All</td><td>100</td></tr><tr><td>shaun2</td><td>4</td><td>All</td><td>0</td></tr><tr><td>sizzlingham</td><td>5</td><td>All</td><td>0</td></tr><tr><td>hello</td><td>6</td><td>All</td><td>0</td></tr><tr><td>gloster</td><td>7</td><td>All</td><td>0</td></tr></tbody></table>	Username	Rank	Worldsection	Points	shaun	1	All	507	freeacc0002	2	All	462	Aangtara	3	All	100	shaun2	4	All	0	sizzlingham	5	All	0	hello	6	All	0	gloster	7	All	0	Leaderboard page ● The Leaderboard page displays the user's rank among other players in the game. ● The Leaderboard Page allows the user to select his/her desired world and section. ● The Show button displays the user's rank and points earned in the selected world and section. ● The Reset button displays the user's rank and total points earned in all worlds.
Username	Rank	Worldsection	Points																														
shaun	1	All	507																														
freeacc0002	2	All	462																														
Aangtara	3	All	100																														
shaun2	4	All	0																														
sizzlingham	5	All	0																														
hello	6	All	0																														
gloster	7	All	0																														
	World Selection page ● The World Selection Page allows the user to select his/her desired world to play in from the choice of: ○ OODP ○ SE ○ SSAD ● The World Selection Page displays the total number of stars earned, and the number of																																

	stars obtained in each world.
	Section Selection page • The section selection page allows the user to select his/her desired section to play in. • The section selection page displays the total number of stars obtained in each section. • The section selection page displays both playable and locked sections.
	PVP World Selection page • The PVP world selection page allows the user to select his/her desired world to play in from the choice of: ○ OODP ○ SE ○ SSAD
	PVP Room Selection page • The PVP room selection page allows the user to select his/her desired room to play in.



Pause page

- The pause page pauses the gameplay.
- The pause page allows the user to access the following features:
 - Menu
 - Save
 - Audio
 - Exit
- Menu allows the user to return to the menu page.
- Save allows the user to save his progress.
- Audio allows the user to change the audio settings of the game.
- Exit allows the user to exit the pause page to resume the game.



Staff User Interface page

- The system will display the available functions.
- The staff can choose to add questions, check student statistics, make an assignment and view leaderboard.

question:

Correct Answer:

Wrong Answers:

Add/Edit Question and Answers page

- The system will prompt the staff to input his/her question, one correct answer and two incorrect answers.

Enter Student: World: Sections:

Name	Stars	Points
freeacc0002	2	282
shaun	4	286
Aanglara	2	100
hello	0	0
gloster	0	0

Statistics page

- The system can display the statistics of students based on the user's selected world and section.
- The staff can search for specific student's statistics.

shaun

Battle stats (PVP)

ODDP		
S1: 286	S2: 0	S3: 0

SE		
S1: 0	S2: 0	S3: 221

SSAD		
S1: 0	S2: 0	S3: 0

Stars (Quest)

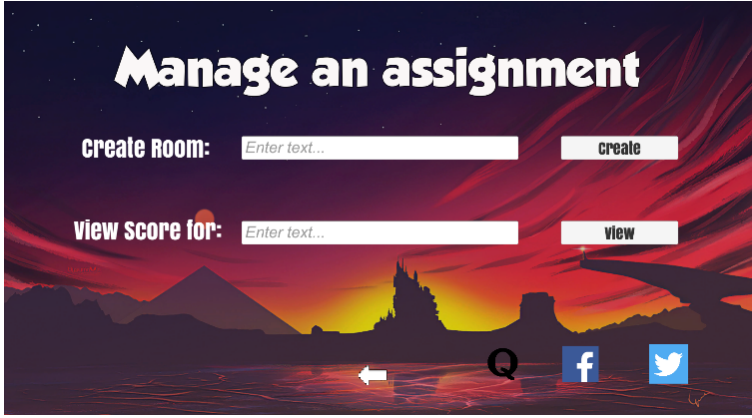
ODDP		
S1: 4	S2: 0	S3: 0

SE		
S1: 0	S2: 0	S3: 0

SSAD		
S1: 0	S2: 0	S3: 0

Students Statistics page

- The system can display the statistics of a specified student.
- The stars and points are broken down to individual worlds and sections.

	Make an Assignment • The system will display 2 input fields on the screen . • The staff can choose to either make an assignment or view score for an assignment. • The staff can also choose to share the assignment on social media via the icons.
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3.2 Hardware Interfaces

The application will require a working laptop/desktop with the minimum software specifications as mentioned in section 2.4 for a smooth user experience. No extra external hardware interfaces will be required.

3.3 Software Interfaces

3.3.1 Firebase Authentication

The user will be prompted to enter their account username and password on the login screen, which will be verified by Firebase Authentication. Users who do not have an existing SoftwareMon account will have to register through the application to create a personal user account for themselves, where the account details will be stored in the firebase.

3.4 Communications Interfaces

The system will use common communication resources, which includes but is not limited to, HTTP protocol to communicate with web browsers.

4. System Features

4.1 Account Creation

4.1.1 Description and Priority

Users will be able to register a new account with Firebase.

Priority for this feature is high as users must register an account in order to use the application.

4.1.2 Functional Requirements

The application must have an interface for new users to register a new user account.

REQ - 1.1	The registration form must request for the new user's personal particulars. • The form must request for the new user's username. • The form must request for the new user's email address. • The form must request for the new user's password. • The form must request for the new user to re-enter his password.
REQ - 1.2:	There must be a "Register" button for users to click on after they have furnished all the details. • The application must be able to verify that all the input fields are filled. • The application must be able to verify that the entered email address is valid. • The application must be able to verify the entered username does not already exist on Firebase. • The application must be able to verify that the entered password is at least 6 characters. • The application must be able to verify that the entered password matches the re-entered password.
REQ - 1.3:	There must be a checkbox for teacher users to toggle an input field. • The input field must request for a teacher pin.

ERROR - 1.4: User has left out one of the required fields to complete the form.	User will not be allowed to proceed with account creation. User will be prompted to input the empty field.
ERROR - 1.5: User has entered an invalid email address	User will not be allowed to proceed with account creation. User will be prompted to input a valid email address.

ERROR - 1.6: User has entered an email address that already exists on Firebase.	User will not be allowed to proceed with account creation. User will be prompted to input a valid email address.
ERROR - 1.7: User has entered a username that already exists on Firebase.	User will not be allowed to proceed with account creation. User will be prompted to input another username.
ERROR - 1.8: User has entered a password of without containing 1 Uppercase, 1 Lowercase and numeric values.	User will not be allowed to proceed with account creation. User will be prompted to input a password of at least 6 characters long.
ERROR - 1.9: User has entered a password that does not match the re-entered password	User will not be allowed to proceed with account creation. User will be prompted to re-enter his password.

4.2 Login

4.2.1 Description and Priority

User logs into his user account by entering his username and password. Priority for this feature is high as every user has to login to the application to be able to use any of the features offered by SoftwareMon.

4.2.2 Functional Requirements

The application must have an interface for users to login.

REQ - 2.1	The login screen must contain input fields for users to input their email address and password.
REQ - 2.2	The password must be encrypted.
REQ - 2.3	There must be an option for a new user to create an account on the login screen.
REQ - 2.4	The login screen must recognise a teacher account and direct the teacher user to the teacher main menu page.

ERROR - 2.5: User has left out either email address or password fields.	User will not be allowed to proceed with login. User will be prompted to input the empty field.
ERROR - 2.6: Username does not exist.	User will not be allowed to proceed with login. User will be prompted to create a new user account.
ERROR - 2.7: Invalid email or password.	User will not be allowed to proceed with login. User will be prompted to check his entered credentials.

4.3 Quest Mode

4.3.1 Description and Priority

The user chooses the quest mode to play the game.

This is one of the primary functionalities of the application, so it has high priority.

4.3.2 Functional Requirements

The application must have a quest mode that consists of 3 worlds for users to learn different phases of the life cycle of Software Engineering.

REQ - 3.1	The quest mode shall have 3 worlds representing the 3 modules related to software engineering. • Object Oriented Design Programming • Software Engineering • Software System Analysis & Design
REQ - 3.2	Each world shall consist of a different theme and background.
REQ - 3.3	Each world shall consist of 3 sections to represent 3-4 topics specific to the world, ranging from basic to advanced.
REQ - 3.4	The quest mode shall have an interface for the user to select a particular world and section.
REQ - 3.5	Each section shall have 4 trainers that users must defeat to proceed to the next section.
REQ - 3.6	Each trainer shall provide users with questions that consist of 3 levels of difficulty, ranging from easy, medium to hard. • The user shall defeat the trainer by reducing its health bar to 0. • The damage dealt to the trainer shall scale with the difficulty of the question. • The trainer shall deal random damage to the user after every turn. • The user shall be defeated by the trainer when his/her health bar reaches 0 before the trainer.
REQ - 3.7	The quest mode shall track the user's progress through the form of stars. • A star shall be awarded to the user whenever a trainer is defeated. • The user shall start with 0 stars in each section, world.

REQ - 3.8	The quest mode shall be replayable. • The user shall be able to challenge defeated trainers again. • The user shall not lose a star if he/she loses to a defeated trainer. • The user shall not gain a star if he/she defeats the defeated trainer again.
REQ - 3.9	User will be allowed to enter any world.

ERROR - 3.9: User has not completed a previous section and wants to enter the next section.	User will not be allowed to enter the next section. User will be prompted to complete the previous section first.
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4.4 PVP Mode

4.4.1 Description and Priority

The user chooses the PVP mode to challenge other users offline.
This is one of the primary functionalities of the application, so it has high priority.

4.4.2 Functional Requirements

The application shall have a PVP mode for users to challenge each other offline.

REQ - 4.1	The PVP mode shall be conducted offline in the form of a challenge room.
REQ - 4.2	The user shall be able to compete in all 3 worlds regardless of his progression in Quest Mode. • Object Oriented Design Programming • Software Engineering • Software System Analysis & Design
REQ - 4.3	Each world shall have 3 challenge rooms, corresponding to the 3 sections.

REQ - 4.4	The PVP mode shall have an interface for the user to select a particular world and challenge room.
REQ - 4.5	Each challenge room shall provide the user with questions from the same question bank that consist of 3 levels of difficulty, ranging from easy, medium to hard. • The challenge shall run on a timer that runs down to 0 from 300 seconds. • The challenge shall provide users with a trainer that has an identical health bar. • The timer will drop by 10 seconds if the user answered a question wrongly. • The user must defeat the trainer by reducing its health bar to 0. • The damage dealt to the trainer shall be relative to the difficulty of the question answered by the user.
REQ - 4.6	The PVP mode shall track the user's proficiency in the form of points. • The user shall be awarded points relative to the time of completion. • The user shall not be awarded any points if he/she fails to defeat the trainer within the assigned 300 seconds.
REQ - 4.7	The PVP mode shall be replayable. • The user shall be able to challenge defeated challenge rooms again. • The user's score for a particular challenge room will be updated if he/she manages to score a higher score than his current score.

4.5 Custom Mode

4.5.1 Description and Priority

The user chooses the custom mode to challenge other users offline with their own set of selected questions. This is one of the primary functionalities of the application, so it has high priority.

4.5.2 Functional Requirements

The application shall have a custom mode for users to challenge each other with their own set of questions.

REQ - 5.1	The custom mode shall be conducted offline in the form of a challenge room.
REQ - 5.2	Each challenge room shall have a unique room ID.

REQ - 5.3	The custom room shall allow users to create a new challenge room or join existing challenge rooms.
REQ - 5.4	The custom mode shall have an interface for the user to create a new challenge room. • The custom mode must request for a new room ID. • The custom mode must verify that the entered room ID does not already exist in Firebase. • The user shall be able to add questions of varying world, section and difficulty from the question bank.
REQ - 5.5	The custom mode shall have an interface for the user to join an existing challenge room. • The custom mode must request for a room ID of an existing challenge room. • The custom mode must verify that the entered room ID exists in Firebase. • The user shall be presented with questions selected by the user who created the challenge room. • The challenge shall end when all the questions added have been answered.
REQ - 5.6	The custom mode shall track the user's proficiency by counting the total number of correct answers. • The user shall be graded by the number of correct answers he/she answers.
REQ - 5.7	The custom mode shall not be replayable. • The user shall not be able to re-enter challenge rooms that he/she has already finished once.

ERROR - 5.8: User has left the required input fields blank.	User will not be allowed to proceed with the user action. User will be prompted to complete the required input field.
ERROR - 5.9: User wants to create a new challenge room with a room ID	User will not be allowed to create a new challenge room. User will be prompted to enter another room ID.

that already exists in Firebase.	
ERROR - 5.10: User wants to enter an existing challenge room with a room ID that does not exist in Firebase.	User will not be allowed to enter the challenge room. User will be prompted to check the room ID of the existing challenge room.

4.6 View Leaderboard

4.6.1 Description and Priority

The user chooses the leaderboard to view the PVP scores of other users.

Priority for this is high as the user has to be able to rank themselves among other users.

4.6.2 Functional Requirements

The application must have a leaderboard interface to display the PVP scores of the users.

REQ - 6.1	The leaderboard shall have an interface that displays all the users in descending order of their points. • The leaderboard shall display the username of every user. • The leaderboard shall display the rank of every user. • The leaderboard shall display the relevant world and section category. • The leaderboard shall display the total points of every user.
REQ - 6.2	The leaderboard shall allow the user to filter the points by world and section.
REQ - 6.3	The leaderboard shall allow the user to see his/her current ranking among all other users.

4.7 Create Assignment

4.7.1 Description and Priority

The teacher user creates an assignment for student users to attempt.
Priority for this is high as the teacher user must be able to assign homework or tasks to their student users to evaluate their proficiency in the module.

4.7.2 Functional Requirements

The application must have an interface for teacher users to create and issue assignments to student users.

REQ - 7.1	The assignment shall be in the form of a challenge room, similar to the custom mode.
REQ - 7.2	The teacher main menu page shall have an interface that allows teacher users to create a new challenge room. • The custom mode must request for a new room ID. • The custom mode must verify that the entered room ID does not already exist in Firebase. • The teacher user shall be able to add questions of varying word, section and difficulty from the question bank.
REQ - 7.3	The teacher main menu page shall have an interface that allows teacher users to share the assignment through social media platforms. • Facebook • Twitter • Quora
REQ - 7.4	The social media post shall include the room ID to the challenge room created by the teacher user.

ERROR - 7.5: Teacher user has left the required input fields blank.	Teacher user will not be allowed to proceed with the user action. Teacher user will be prompted to complete the required input field.
ERROR - 7.6: Teacher user wants to create a new challenge room with a room ID that already exists in Firebase.	Teacher user will not be allowed to create a new challenge room. Teacher user will be prompted to enter another room ID.

4.8 View Assignment Score

4.8.1 Description and Priority

The teacher user views the scores of his/her students who have attempted the assignment.

Priority for this is high as the teacher user must be able to evaluate his/her students' proficiency in the module.

4.8.2 Functional Requirements

The application must have an interface for teacher users to view the scores of student users who have attempted the assignment.

REQ - 8.1	The teacher main menu page shall have an interface that allows teacher users to enter the challenge room they have created. • The custom mode must request for a room ID for an existing challenge room. • The custom mode must verify that the entered room ID exists in Firebase.
REQ - 8.2	The challenge room shall display the list of student users who have attempted the assignment and their respective scores. • The challenge room shall display the username of every student user. • The challenge room shall display the number of correct answers answered by every student user.

ERROR - 8.3: Teacher user has left the required input fields blank.	Teacher user will not be allowed to enter the challenge room. Teacher user will be prompted to complete the required input field.
ERROR - 8.4: Teacher user wants to enter an existing challenge room with a room ID that does not exist in Firebase.	Teacher user will not be allowed to enter the challenge room. Teacher user will be prompted to check the room ID of the challenge room they have created.

4.9 View Discussion Forum

4.9.1 Description and Priority

Student and teacher users discuss questions and answers on an online discussion forum.
Priority for this is medium as this is not the main functionality of the application.

4.9.2 Functional Requirements

The application shall provide a discussion forum for teacher and student users to participate in discussions.

REQ - 9.1	The application shall have an interface that allows users to access the discussion forum through a URL link.
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4.10 View Statistics

4.10.1 Description and Priority

Users view their own statistics.
Priority for this is high as users need to be able to track their personal progress in the game.

4.10.2 Functional Requirements

The application shall have an interface for teacher users to track the statistics of student users.

REQ - 10.1	The main menu page shall have an interface for the user to view his/her own statistics. • The main menu page shall display the username of the logged in user. • The main menu page shall display the total stars of the logged in user. • The main menu page shall display the total points of the logged in user.
REQ - 10.2	The main menu page shall allow the user to filter the points by world and section.

4.11 View Student Statistics

4.11.1 Description and Priority

Teacher users view their students' statistics.

Priority for this is high as teacher users need to be able to track their students' personal progress in the game.

4.11.2 Functional Requirements

The application shall have an interface for teacher users to track the statistics of student users.

REQ - 11.1	The teacher main menu page shall have an interface for the teacher user to access each student's statistics by entering their username.
REQ - 11.2	The teacher main menu page shall display the specified student's statistics. • The teacher main menu page shall display the username of the specified student user. • The teacher main menu page shall display the total stars of the specified student user. • The teacher main menu page shall display the total points of the specified student user.
REQ - 11.3	The main menu page shall allow the teacher user to filter the statistics by world and section.

4.12 Manage Question Bank

4.12.1 Description and Priority

Teacher users to be able to add, edit questions in the existing database.

Priority for this is high as teacher users need to be able to monitor the questions in the database.

4.12.2 Functional Requirements

The application shall have an interface for teacher users to track the statistics of student users.

REQ - 12.1	The teacher main menu page shall have an interface for the teacher user to access add or edit questions.
REQ - 12.2	The question bank shall be segmented into 3 worlds with 3 sections. • Each section shall have questions of varying difficulties from easy to hard.

REQ - 12.3	The teacher shall be able to add questions into the question bank. • The system shall present a series of dropdown options for the teacher to select the topic (world), section and difficulty. • The teacher shall proceed to input the question and answers field. • The question and answer options shall be added into the database based on the teacher's input.
REQ - 12.4	The teacher shall be able to modify an existing question. • The system shall present a filter form for staff to select the section and difficulty. • The questions fulfilling this criteria shall be listed. • The teacher shall be able to select questions from the list to be modified. • The question and answers of existing question will populate a form for the staff to amend. • The teacher will submit the amended question and answers which will change the existing question and answer in the database.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

1. The application must be able to register a new user successfully within **5 seconds**.
2. The application must be able to log in successfully within **5 seconds**.
3. The application must display the correct interface within **5 seconds** of user input.
4. The application must be able to utilise any combination of the filter categories and display the desired search results within **5 seconds** of user input.

5.2 Safety Requirements

The application does not have an external device that may harm the life of the user. However, users are advised to exercise discipline when playing the game to avoid health related incidents occurring from excessive gaming.

5.3 Security Requirements

Sensitive information such as the user's account login information must not be sent anywhere except onto the application and the application's Firebase database. Students and teachers alike have to set a password that contains at least 1 Uppercase, 1 Lowercase and a numeric number. Passwords will be encrypted as well. Student accounts cannot be allowed to use specialised functions developed for teachers and admins. Student is also not allowed to register as a teacher as the pin number to register as a student is only made known to the teacher. The use of school email accounts in account creation prevents players from holding more than 1 account at a time.

5.4 Software Quality Attributes

Usability

1. The application must be able to reduce short-term memory retention. The simple design of the game aids in **reducing memory load**, as all features in the game are easily accessible without having to memorise any steps.
2. A maximum of xxx characters will be displayed on the page during battle, reducing cluttering and enhancing **readability for users**.
3. The system strives to **cater to universal usability** through the use of intuitive icons such as a lock to denote that a feature is not accessible, or an arrow pointing to the left to indicate the back button.
4. The application allows for **effortless reversal of any actions**. Users are able to go back to their previous page or return to the main menu from any page in the application.
5. **95% of users** must know how to navigate through the application to access the different game modes.
6. The application must have a consistent visual layout throughout the various pages. The back button on the application must remain in the same (top right) location on every page of the application. The color scheme must also be consistent on every page of the system.

Reliability

1. Once the application is launched, it must observe **full system functionality**.

2. All changes made within the application **must be saved** when the application is suspended or closed.
3. The application must not crash **95% of the time**.

Learnability

1. **90% of first time users** should be able to navigate through the application and access the quest mode **within 5 minutes of beginning to use the application**.

Extensibility

1. The application must be designed with the **Entity-Controller-Boundary** architecture and design patterns to support any future enhancements and facilitate extendability.

Supportability

1. The application is a Unity project and must be supported on any Mac OS or Windows device.
2. The backend code of the application is written entirely in C#.

Reusability

1. Our game design has decomposed the system into reusable components such as the BattleScene, Leaderboard via abstraction.

5.5 Business Rules

Our application does not have any business rules pertaining to any specific user group.

6. Other Requirements

Appendix A: Glossary

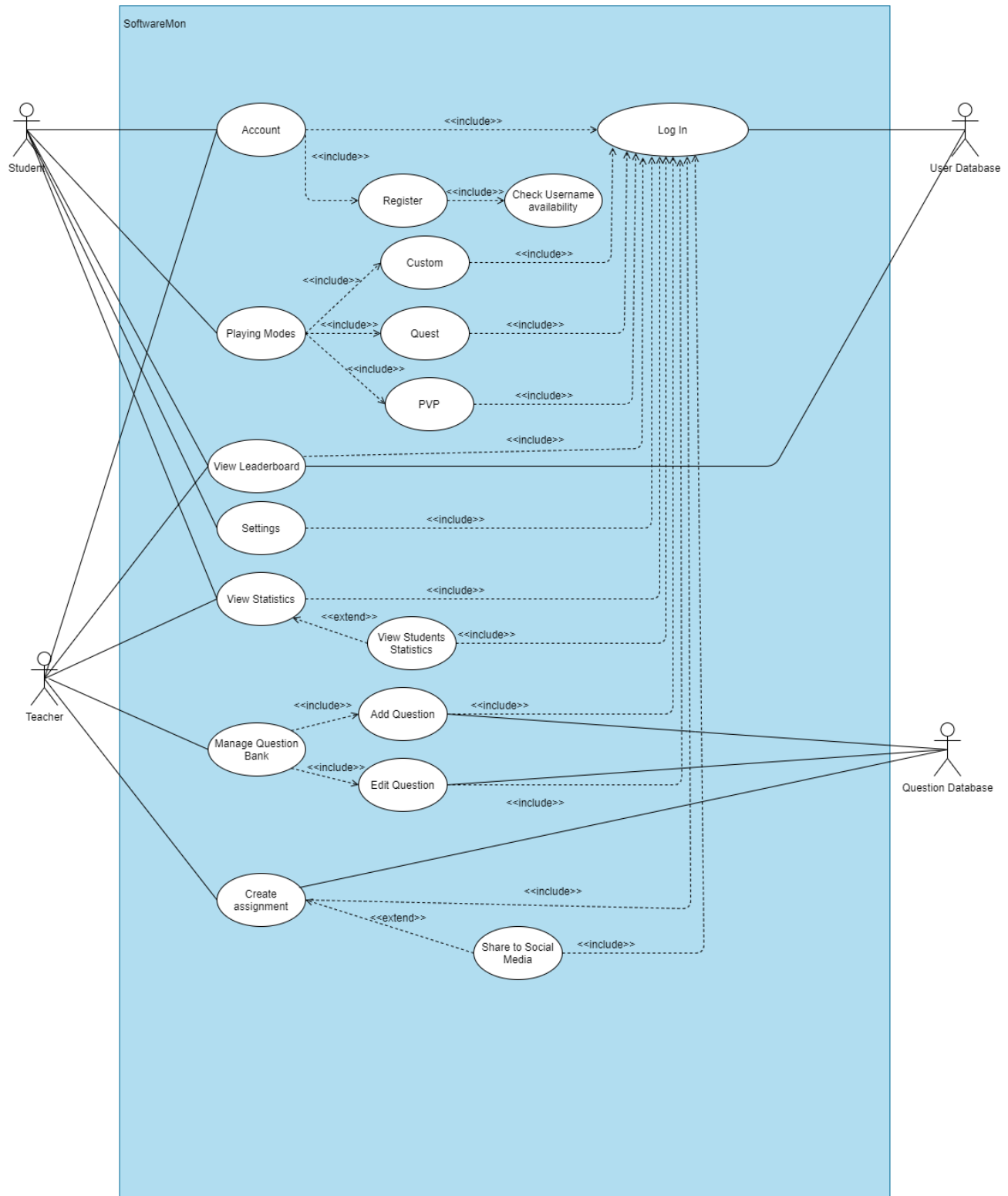
Data Dictionary

Term	Definition
User	Users refer to the various focus groups of our application, specifically: - Students - Teachers Students refer to students in NTU under the SCSE curriculum Teachers refer to the teaching staff in NTU employed under the SCSE curriculum.
Stars	Stars refer to the metric which our application uses to track the progress of users in the quest mode.
Points	Points refer to the metric which our application uses to rank users in the PVP mode.
OODP	OODP (Object Oriented Design & Programming) refers to the computing module provided at Nanyang Technological University.
SE	SE (Software Engineering) refers to the computing module provided at Nanyang Technological University.
SSAD	SSAD (Software Systems Analysis & Design) refers to the computing module provided at Nanyang Technological University.
World	A world represents the different phases of Software Engineering modules. • OODP • SE • SSAD
Section	A section is a subunit of a world and represents the different topics of the module in the world.
Trainer	A trainer is a subunit of a section and represents the different stages the user has to defeat in a section. The user can enter a battle with the trainer.

Difficulty	Difficulty refers to the levels of difficulty that are used to segment the questions in the question bank.
Question Bank	The question bank refers to the full list of questions that are used in battle.
Battle	In a battle, the user has to answer a series of questions.
Leaderboard	The leaderboard is a scoreboard showing the names and current scores of the leading competitors
Social Media	Social media is an online platform where users can post online.

Appendix B: Analysis Models

6.1 Use Case Model



6.1.1 Use Case Descriptions

1 Account Creation

Use Case ID:	UC001
Use Case Name:	Create Account

Actor:	Student/Staff
Description:	The user can create his/her account.
Preconditions:	The user does not have an existing account.
Postconditions:	The user has created his/her account.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. The user clicks on “Register”. 2. The user indicates if he/she is a “Student” or “Teacher”. 3. The user indicates his/her preferred username, email address and password. 4. The user will be required to re-enter password as confirmation. 5. The user clicks on “Register” to finish creating the account. 6. A message “Account Created Successfully” is

	displayed on the screen.
Alternative Flows:	<u>AF-S2:</u> If the user indicates Teacher 1. The system will require the Teacher to key in the pin “PIN1” (only known to the teacher) so that other users would not freely register as a Teacher. 2. The system displays “Invalid Pin!” if incorrect Pin is used.
	<u>AF-S5-1:</u> If the email address is already in use: 1. The system displays “Email already in use”. 2. The system returns to step 3. <u>AF-S5-2:</u> If the username is in used: 1. The system displays “Username taken!”. 2. The system returns to step 3. <u>AF-S5-3:</u> If the password is lesser than 6 characters: 1. The system displays “Weak password”. 2. The system returns to step 3. <u>AF-S5-4:</u> If the passwords do not match: 3. The system displays “Password does not match”. 4. The system returns to step 3. <u>AF-S5-5:</u> If any of the input is missing: 1. The system displays “Missing input”, depending on which field is missing. 2. The system returns to step 3.
Exceptions:	-

Includes:	-
Special Requirements:	-
Assumptions:	-
Notes and Issues:	-

2 Login

Use Case ID:	UC002
Use Case Name:	Log In

Actor:	Student/Staff
Description:	Student/Staff logs in to the game system via a registered email address. This use case is a precondition for all the other use cases.
Preconditions:	The user has successfully started the application.
Postconditions:	The user has logged in to the game system with an account.
Priority:	High
Frequency of Use:	High

Flow of Events:	1. The user opens the application. 2. The system displays the login page. 3. The user enters his/her username and password. 4. The user clicks the login button 5. The user successfully logs in to the system. 6. A message “Log in Successfully” is displayed on the screen.
Alternative Flows:	<u>AF-S4-1:</u> If the authentication fails: 1. The system displays “Incorrect password” or “Invalid email”, depending on which field is invalid. 2. The system returns to Step 2. <u>AF-S4-2:</u> If the email or password field is missing: 1. The system displays “Missing email” or “Missing Password” respectively. 2. The system returns to Step 2. <u>AF-S4-3:</u> If the user account does not exist: 1. The system displays “Account does not exist”. 2. The system returns to Step 2. 3. The user will have to register for an account.
	<u>AF-S3:</u> If they forgot password: 1. The user clicks on “Forgot Password”. 2. The system displays the email verification page. 3. The user enters their NTU email address into the text field. 4. The system checks the email validity, and displays email sent. 5. The user will go to their inbox to reset his/her

	password.
	<u>AF-S6-1</u>: If the user account is a student: 1. The student user will be directed to the main menu page <u>AF-S6-2</u>: If the user account is a staff: 1. The teacher user will be directed to their own teacher main menu page
Exceptions:	-
Includes:	-
Special Requirements:	-
Assumptions:	-
Notes and Issues:	-

3 Quest Mode

Use Case ID:	UC006		
Use Case Name:	Play Quest Mode		
Created By:		Last Updated By:	
Date Created:	13 Feb 2021	Date Last Updated:	

Actor:	Student
Description:	The user chooses the Quest Mode to play the game.

Preconditions:	The user has logged in.
Postconditions:	-
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user chooses Quest Mode. 2. The system displays 3 different world maps to be explored in the Quest Mode. 3. The system displays the total number of stars obtained from all worlds. 4. The user chooses a world to enter by clicking on it. If the user is playing the world for the first time, the system displays the number of stars earned in that world to be zero. 5. The user will choose the section that he/she wants to play in. If the user is playing the section for the first time, the system displays the number of stars earned in that section to be zero. 6. Each section will consist of 3-4 topics from each module. 7. For each section, there will be 4 enemy trainer NPCs scattered across the map. 8. For each section, the user will have to find the enemy trainer NPCs and defeat them. 9. If a user walks in front of the enemy trainer NPC for the first time, it will automatically walk towards the user and challenge him/her. The scene will change to a battle scene. 10. Both the user and the enemy trainer will be given a

	100 health points at every start of a battle. 11. The user will be given an option to fight or run from the battle throughout the duration of the battle. 12. The user will choose either Easy, Medium or Hard questions and each correct answer will deal damage to the enemy trainer corresponding to its difficulty level. 13. The enemy trainer will deal damage of random move (Easy, Medium or Hard) to the user. 14. Once the enemy's health points reach zero, the user will earn a star from the battle. The user can only earn a star from a trainer once. 15. The user has to obtain a total of 4 stars to unlock the next section.
Alternative Flows:	<u>AF-S4-1: If the user has attempted the world previously:</u> 1. The user shall be able to see the total number of stars earned in that world which is the sum of the stars earned in all 3 sections in that world. <u>AF-S5-1: If the user has completed a section in the world previously:</u> 1. The user shall see that the lock icon on the next section will disappear and they will be able to click on it. <u>AF-S5-2: If the user has attempted but not yet completed a section in the world previously:</u> 1. The user shall be able to see the number of stars

	he/she earned in that section. <u>AF-S5-2:</u> If the user wants to try the same section again: 1. The user chooses to replay the section he has completed. 2. The system shall allow the user to continue fighting trainers. 3. The user's star count will not increase any further. <u>AF-S10:</u> If the user's chooses run: 1. The system returns to step 8 and the user will not be awarded a star from the trainer. <u>AF-S11-1:</u> If the user's health points reached 0 before the enemy trainer: 1. The user will faint and lose the battle. 2. The system returns to step 8. 3. The student can re-try this section and battle the trainers again. 4. The student will not lose stars despite losing the battle. <u>AF-S11-2:</u> If the user answered the question wrongly: 1. The user will skip a turn and the enemy trainer will deal damage of random move (Easy, Medium or Hard) to the user. 2.
Exceptions:	-
Includes:	Log In

Special Requirements:	-
Assumptions:	-
Notes and Issues:	If the student plays the same section again which he/she has already completed, the new attempt does not affect his/her star count for the section.

4 PVP Mode

Use Case ID:	UC007		
Use Case Name:	Play PVP Mode		
Created By:		Last Updated By:	
Date Created:	13 Feb 2021	Date Last Updated:	

Actor:	Student
Description:	The user chooses the PVP Mode to play the game.
Preconditions:	The user has logged in.
Postconditions:	-
Priority:	High
Frequency of Use:	High

Flow of Events:	1. The user selects PVP mode. 2. The user selects the desired world to be played in. 3. The user selects the room which is equivalent to the section. 4. The user will be transported to the battle scene and he/she will battle an enemy trainer. 5. The user is given 300 seconds to complete the battle. 6. The user shall choose either Easy, Medium or Hard questions and each correct answer will deal damage to the enemy trainer corresponding to its difficulty level. 7. A timer will be displayed to indicate the remaining time to defeat the trainer. 8. The timer will only runs when the player is answering the question and not when the question is being printed out on display. 9. Once the trainer health points decreases to zero, the user will earn points which takes the remaining time rounded of the nearest seconds. 10. The system will only record the highest points attained from the user in that room onto the firebase database.
Alternative Flows:	<u>AF-S5-2:</u> If the user answers the question incorrectly: 1. He/She will do 0 damage to the opponent. 2. He/she will be deducted 10 seconds from the timer count. <u>AF-S7:</u> If the user runs out of time in the battle: 1. He/She will gain no points from the battle.

Exceptions:	-
Includes:	Log In
Special Requirements:	-
Assumptions:	-
Notes and Issues:	If the user plays the same section again which he/she has already completed and scored points lower than his/her highest recorded points, the attempt shall not affect his/her points gained for the section.

5 Custom Mode

Use Case ID:	UC008		
Use Case Name:	Play Custom Mode		
Created By:		Last Updated By:	
Date Created:	13 Feb 2021	Date Last Updated:	

Actor:	Student/Staff
Description:	The user chooses the custom mode to play the game.
Preconditions:	The user has logged into the system with his NTU account.
Postconditions:	-
Priority:	Medium
Frequency of Use:	Medium

Flow of Events:	1. The user chooses custom mode. 2. The user will be given the option to create room or enter a room. 3. The user enters the battle scene after keying in a valid roomID. 4. The user will be dealt with questions set by another user or teacher in order. 5. The user will answer each questions every round after he/she chose “Fight”. 6. The user will not receive any penalty if the player answers a question wrongly. 7. Once the user finished answering all questions, the system will notify the student of their score before terminating the battle scene. 8. The score will be saved onto the firebase database along with the user’s username. 9. Only the teacher will be able to view the scores of each student that has attempted the custom game that he/she created.
Alternative Flows:	<u>AF-S2-1:</u> If the user chooses to create a room: 1. The user will first be required to enter a roomID that he/she wants to create. 2. The user will receive a warning text that says “Room already exist!” if the entered roomID is being used already. 3. The user will be presented with a screen to filter the questions to add to the assignment. Filters include a. World b. Section c. Difficulty

	- After selecting the filters, the user will be presented with the questions from that world, section and difficulty - Upon clicking add for the question that is selected, the question is added to the assignment. - The user will not be able to add the unique questions into the custom room more than once. - The questions will be added to the temporary database created for each room. <u>AF-S3-1:</u> If the user enters a roomID that does not exist: - The user will receive a warning text that says “RoomID does not exist!” <u>AF-S3-2:</u> If the user enters a roomID that he/she has previously attempted: - The user will receive a warning text that says “You have attempted already!” <u>AF-S5-1:</u> If the user chose “Run”: - The user will be transported back to the game mode page.
Exceptions:	-
Includes:	Log In
Special Requirements:	-
Assumptions:	-
Notes and Issues:	-

6 View Leaderboard

Use Case ID:	UC010		
Use Case Name:	View Leaderboard		
Created By:		Last Updated By:	
Date Created:	13 Feb 2021	Date Last Updated:	

Actor:	Student/Staff
Description:	The user can view their rankings based on points obtained in PVP and compare their rankings with other players.
Preconditions:	The user has logged in to the system.
Postconditions:	The user has learnt about their positions or other players' positions .
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The user clicks on the “View Leaderboard” button. 2. The system will display the top students based on the selected world and section. 3. The system will display the user’s ranking and points obtained in the selected world and section.
Alternative Flows:	<u>AF-S1:</u> If the user is a staff: 1. The staff will not be able to see his/her own ranking as they are not assigned a point rating.

Exceptions:	-
Includes:	Log In

Special Requirements:	-
Assumptions:	-
Notes and Issues:	-

7 Create Assignment

Use Case ID:	UC0011		
Use Case Name:	Create Assignment		
Created By:		Last Updated By:	
Date Created:	13 Feb 2021	Date Last Updated:	

Actor:	Staff
Description:	The user can create assignments which can be shared to students via social media.
Preconditions:	The user has logged in to the system.
Postconditions:	The assignment is created The assignment can be shared on social media
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The user clicks on the “Make an Assignment” button. 2. The user can select 2 options a. Create Room b. View Score for Room 3. If the user selects to create a room, he/ she will be presented with a screen to filter the questions to add to

	the assignment. Filters include - World - Section - Difficulty - After selecting the filters, the user will be presented with the questions from that world, section and difficulty - Upon clicking add for the question that is selected, the question is added to the assignment. - The user will not be able to add the unique questions into the custom room more than once. - After selecting all the questions for the assignment, the user clicks on the finish button. - The user is brought back to the page where he/she can create or view scores for the room. - The user chooses which social media platform to share the room ID and password: - Facebook - Twitter - Quora
Alternative Flows:	AF-S2a - If the room already exist, an error message “Room already exist!” is displayed - Return to step 2 AF-S2a - If the field is empty, an error message “Please input something” is displayed - Return to step 2 AF-S2b - If the room does not exist, an error message “Room

	does not exist” is displayed 2. Return to step 2 AF-S2b 1. If the room is valid, the user will be presented with a list of users who have attempted the assignment and their respective score
Exceptions:	-
Includes:	Log In
Special Requirements:	-
Assumptions:	-
Notes and Issues:	-

8 View Statistics

Use Case ID:	UC009		
Use Case Name:	View Statistics		
Created By:		Last Updated By:	
Date Created:	13 Feb 2021	Date Last Updated:	

Actor:	Staff, Student
Description:	The user can find out about their statistics. The teacher can find out about the student's statistics.

	Statistics include: 1. Username 2. Stars of the player 3. Points of the Player
Preconditions:	The user has logged in to the system.
Postconditions:	The staff or student has learnt about the statistics.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The student login to the game and accesses the main menu. 2. The user can select the filter of world and section. 3. The System will display the students statistics. Statistics include: 1. Username 2. Stars of the player 3. Points of the player 4. Total stars of the player 5. Total points of the player
Alternative Flows:	<u>AF-S2</u>: If the user is a Staff: 1. The names of students will show up in a scrollable list with their stars and points first. 2. Staff can alter the world and section filter to obtain the data for the respective world or section. 3. The staff will search for the student through a search bar. 4. The flow follows step 2 after this.

Exceptions:	-
Includes:	Log In
Special Requirements:	-
Assumptions:	-
Notes and Issues:	-

9 Manage Question Bank

Use Case ID:	UC005		
Use Case Name:	Manage Question Bank		
Created By:		Last Updated By:	
Date Created:	13 Feb 2021	Date Last Updated:	

Actor:	Staff
Description:	The user can amend the question bank via 2 ways: Adding or editing a new question.
Preconditions:	The user has logged in to the system.
Postconditions:	The question bank in the database has been updated after the new amendments.

Priority:	Low
Frequency of Use:	Low
Flow of Events:	1. The user shall be able to add new questions. 2. The system presents drop down options that allows users to select the topic (world), section and difficulty. 3. The user clicks the “Next” button. 4. The system presents an entry form to the user. The required fields to enter include the question, one correct answer and two wrong ones. 5. The user user clicks the “Submit” button. 6. The system processes the submission and updates the question bank in the database. 7. The user shall be able to edit existing questions and answers via another screen.
Alternative Flows:	<u>AF-S5:</u> If the user does not provide any inputs for questions and answers: 1. The system displays an error message “Missing Input(s)” depending on which field is missing. 2. The question bank remains the same. AF-S1 If the user chooses to edit a question 1. The system presents drop down options that allows users to select the topic (world), section and difficulty. 2. The user clicks the “Next” button. 3. The system presents an entry form to the user. The existing question and answers will populate the form. 4. The user amends the questions and/or answers. 5. The user user clicks the “Submit” button. 6. The system processes the submission and updates the

	question bank in the database.
Exceptions:	-
Includes:	Log In
Special Requirements:	-
Assumptions:	-
Notes and Issues:	-

10 Settings

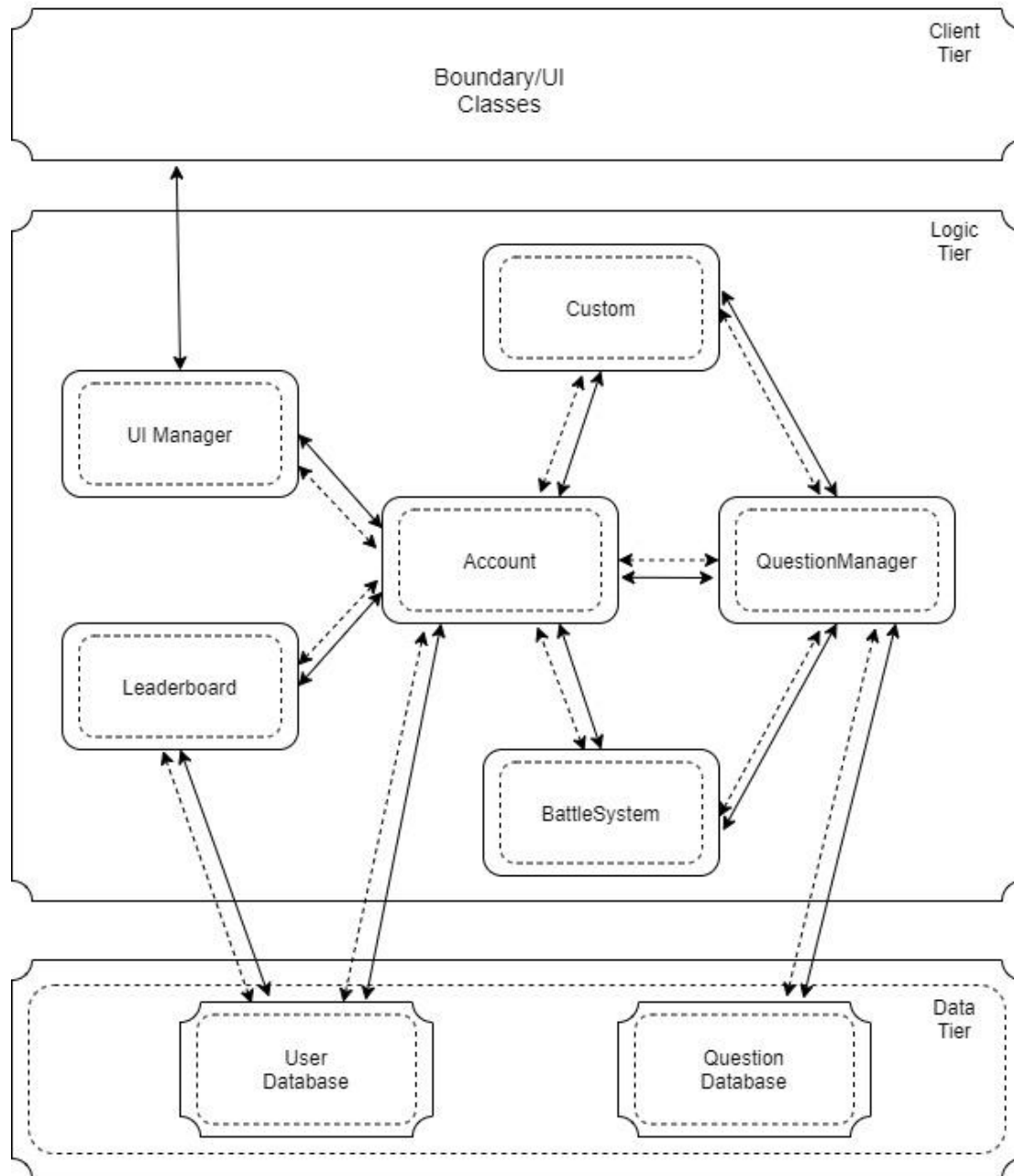
Use Case ID:	UC004		
Use Case Name:	Setting		
Created By:		Last Updated By:	Cheong Jun Hao
Date Created:	13 Feb 2021	Date Last Updated:	6 April 2021

Actor:	Student
Description:	The user can edit his/her preferences through the settings in the game system.
Preconditions:	The user has logged in to the system. The user has clicked escape button while in the quest mode
Postconditions:	The new settings have been updated in the system.

Priority:	Low
Frequency of Use:	Low
Flow of Events:	1. The user accesses the settings screen. 2. The user changes his audio settings. 3. The user presses the “Menu” button. 4. The user presses the “Save” button. 5. The user presses the “Exit” button.
Alternative Flows:	<u>AF-S3-1:</u> If the user presses the “Menu” button: 1. The system will quit the game and change its scene back to the Game Mode page. <u>AF-S4-1:</u> If the user presses the “Save” button: 1. The system will save the player’s stars earned onto the online firebase database. <u>AF-S5-1:</u> If the user presses the “Exit” button: 1. The system will log the player out of the game and change its scene back to the Log in page.
Exceptions:	-
Includes:	Log In
Special Requirements:	-
Assumptions:	-
Notes and Issues:	-

6.2 Candidate Architecture

3-tier architecture

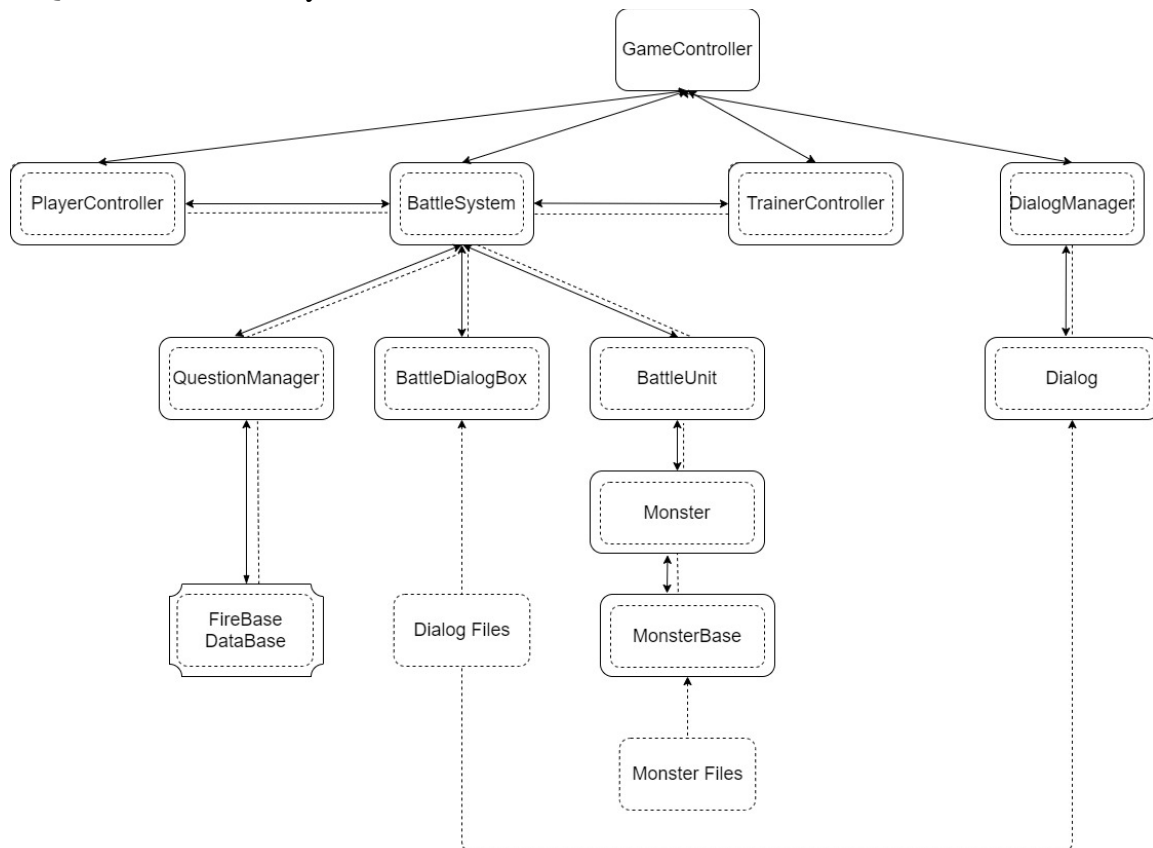


We have chosen the 3 tier architecture. It contains the client tier, logic tier and the data tier. The client tier contains all of the boundary and UI classes, this is where the users are supposed to command and control the application. The next tier is the logic tier, this is where the control functions are stored. It is executed autonomously or by human command. The final tier is the data tier where persistent data is stored. This includes our question database and user database. This

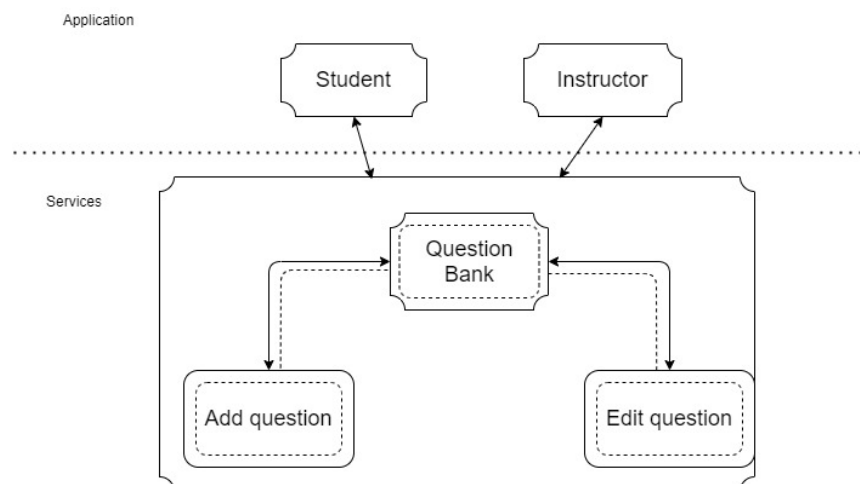
architecture fits our system as the data is changed as the user plays the game, but the presentation needs to reflect the changes in the data. As such the presentation layer remains the same for each of the different UI, while pulling the new data from the data tier through the logic tier.

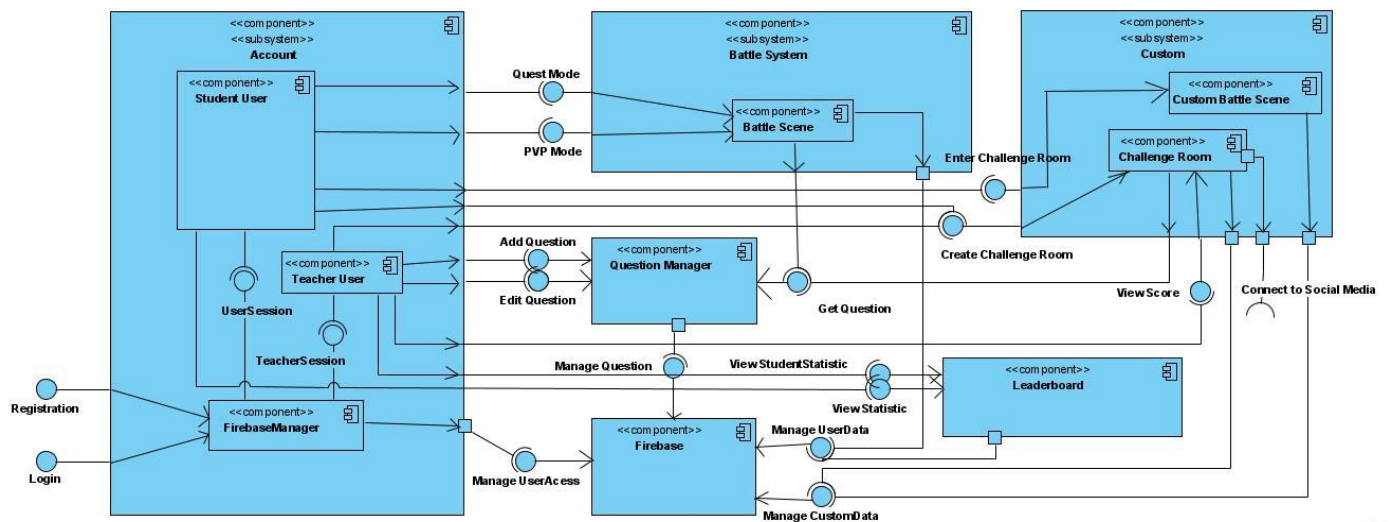
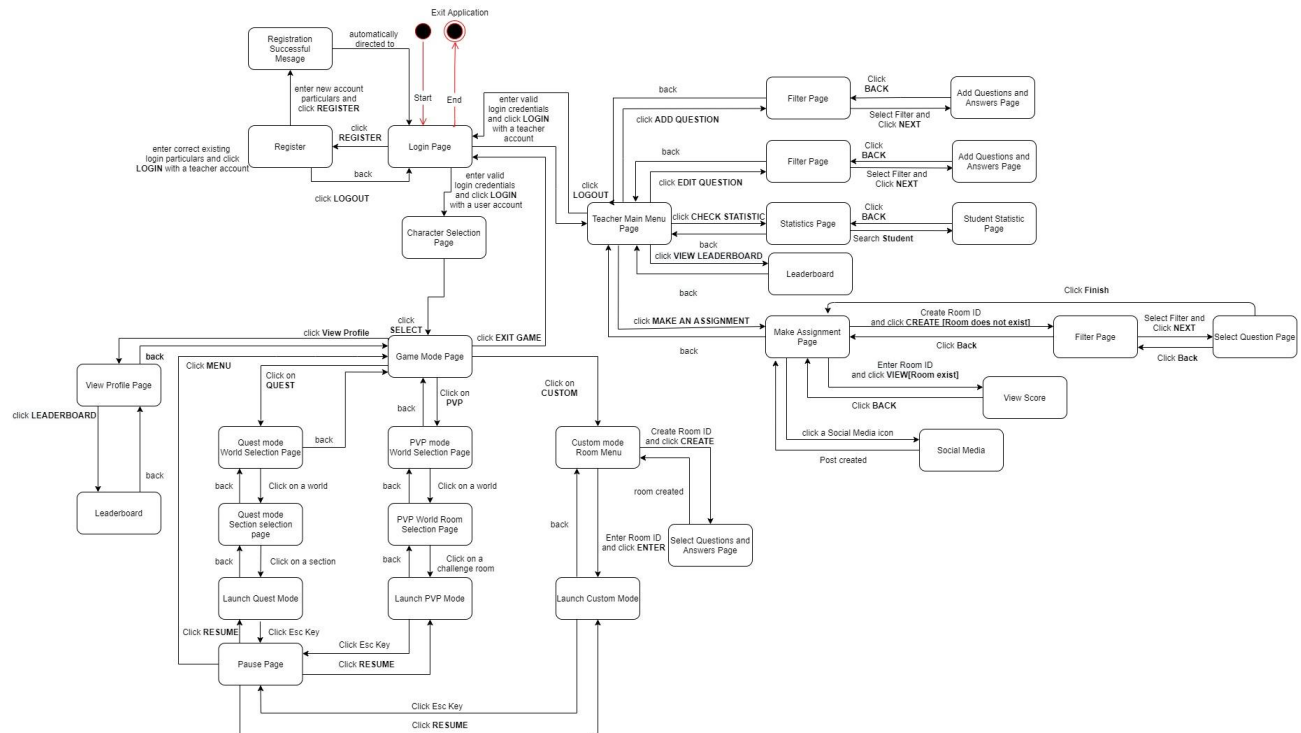
6.3 Subsystem Interface Design

6.3.1 Quest Mode Battle System



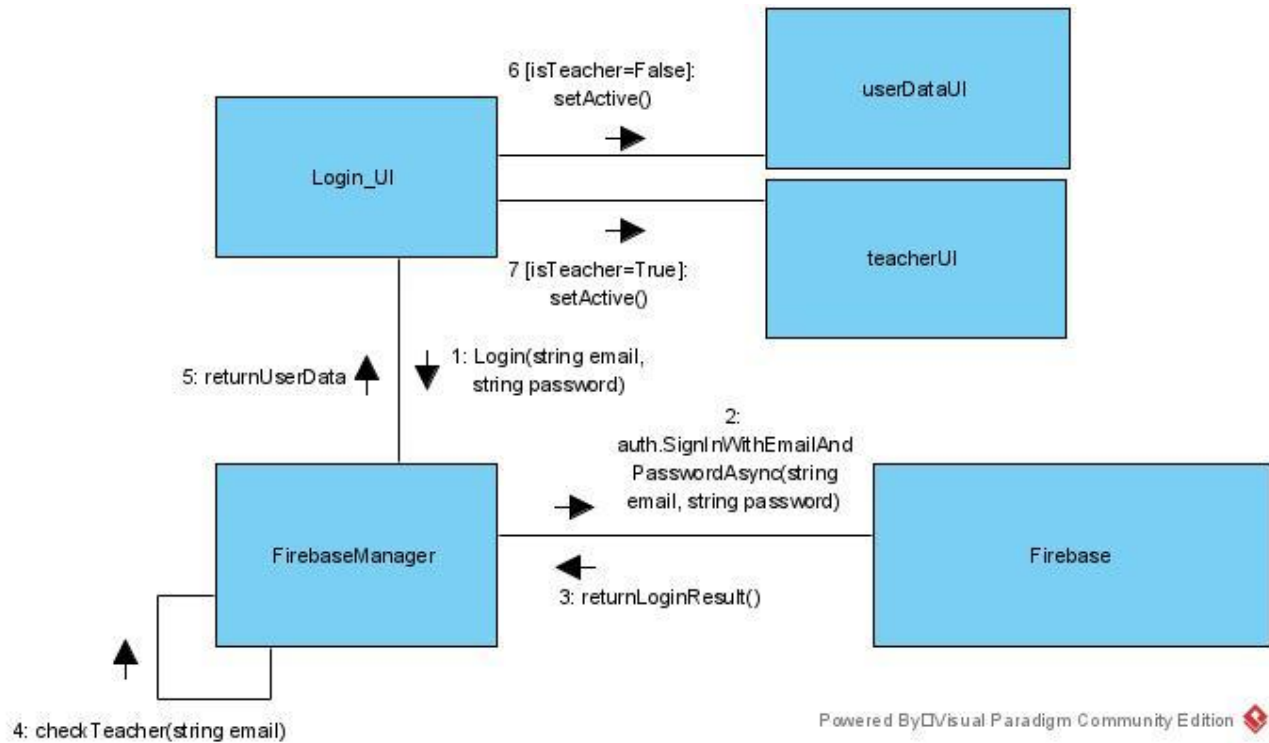
6.3.2 Question Bank Interface Design



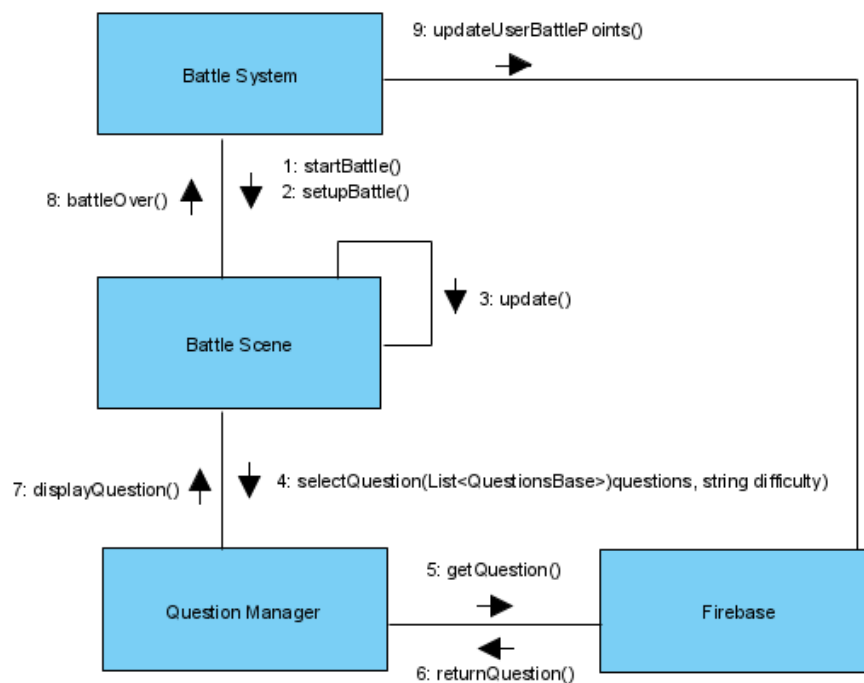


6.6 Communication Diagram

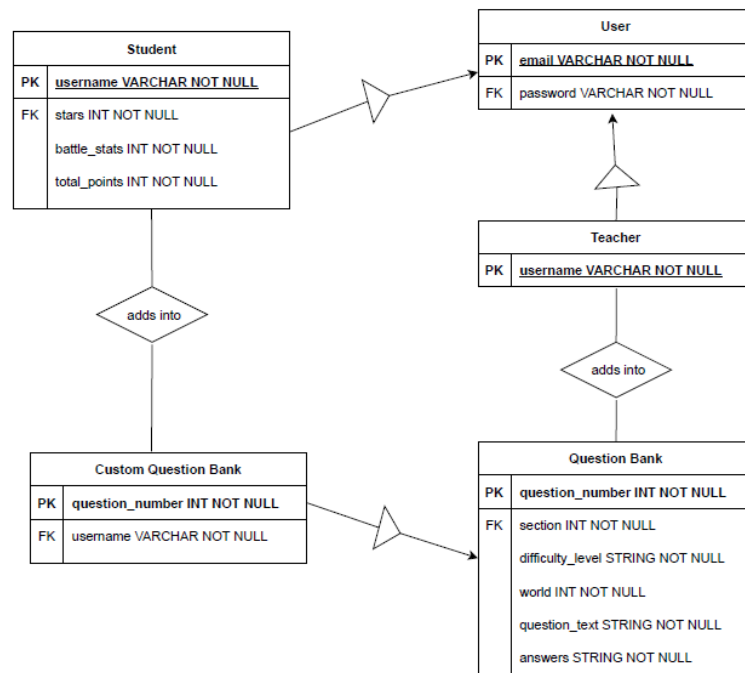
6.6.1 Login



6.6.2 BattleSystem



6.7 ER Diagram



Appendix C: Testing

7. Black-Box Testing

7.1 Account Creation

Equivalence Class

Username	
New Username	New username
Existing Username	Username already taken
Email	
Valid Email	Strings containing character @ and .com
Invalid Email	Strings not containing character @ and .com
Invalid Email	Email already in use
Password	
Valid Password	At least 6 characters long with 1 uppercase letter, 1 lowercase letter and a number
Invalid Password	Less than 6 characters/ Not entered
Invalid Password	Does not include uppercase letter, lowercase letter and number
Confirm Password	
Valid Confirm Password	Matches Password
Invalid Confirm Password	Does not match Password
Teacher	
Correct Unique Pin	Pin = "PIN1"
Incorrect Unique Pin	Invalid Pin

Student Account:

Case no.	Username	Email	Password	Confirm Password	Expected Output	Actual Output
1	Entered	Valid Email	Valid	Match	Registration	Registration

			Password	Password	Successful	Successful
2	Existing Username	Valid Email	Valid Password	Match Password	Registration Unsuccessful	Registration Unsuccessful
3	Not Entered	Valid Email	Valid Password	Match Password	Registration Unsuccessful	Registration Unsuccessful
4	Entered	Not Entered	Valid Password	Match Password	Registration Unsuccessful	Registration Unsuccessful
5	Entered	Invalid Email Format	Valid Password	Match Password	Registration Unsuccessful	Registration Unsuccessful
6	Entered	Email in Use	Valid Password	Match Password	Registration Unsuccessful	Registration Unsuccessful
7	Entered	Valid Email	Not Entered/ <6 characters	Match Password	Registration Unsuccessful	Registration Unsuccessful
8	Entered	Valid Email	Password does not contain upper and lowercase letter	Match Password	Registration Unsuccessful	Registration Unsuccessful
9	Entered	Valid Email	Valid Password	Not Entered	Registration Unsuccessful	Registration Unsuccessful
10	Entered	Valid Email	Valid Password	Does not match Password	Registration Unsuccessful	Registration Unsuccessful

Teacher Account:

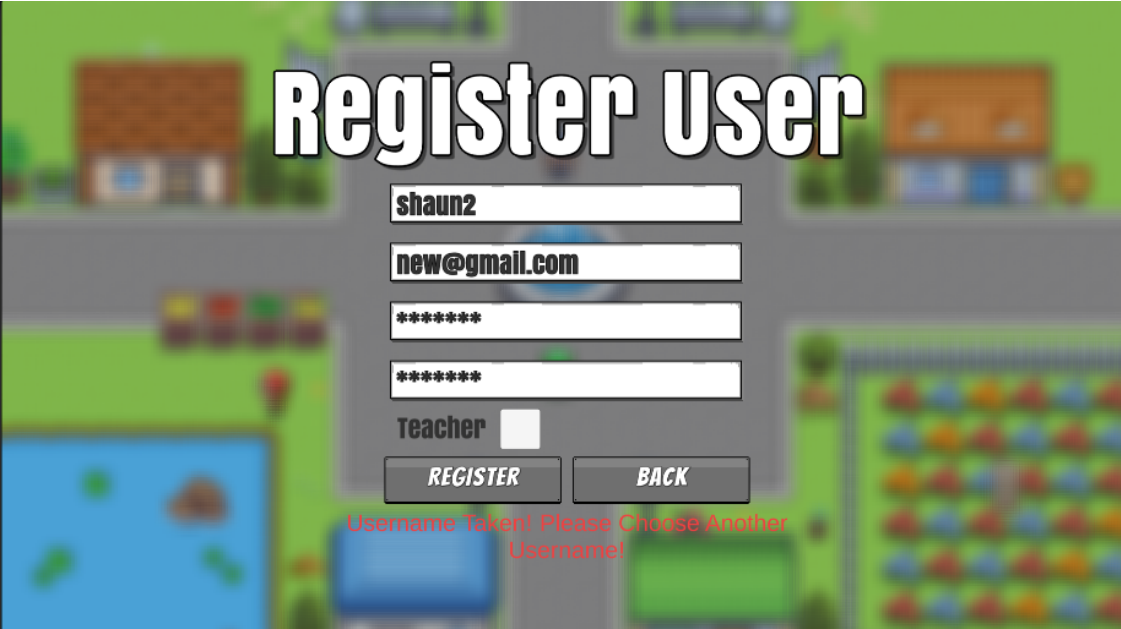
Checking of inputs are exactly the same as that of a student account with an additional Unique Pin input

Case no.	Username, Email, Password, Confirm Password	Teacher Checkbox	Unique Pin	Expected Output	Actual Output
11	Valid Username, Email, Password, Confirm	Checked	Valid Pin	Registration Successful	Registration Successful

	Password				
12	Valid Username, Email, Password, Confirm Password	Checked	Not Entered	Registration Successful	Registration Successful
13	Valid Username, Email, Password, Confirm Password	Checked	Invalid Pin	Registration Unsuccessful	Registration Unsuccessful

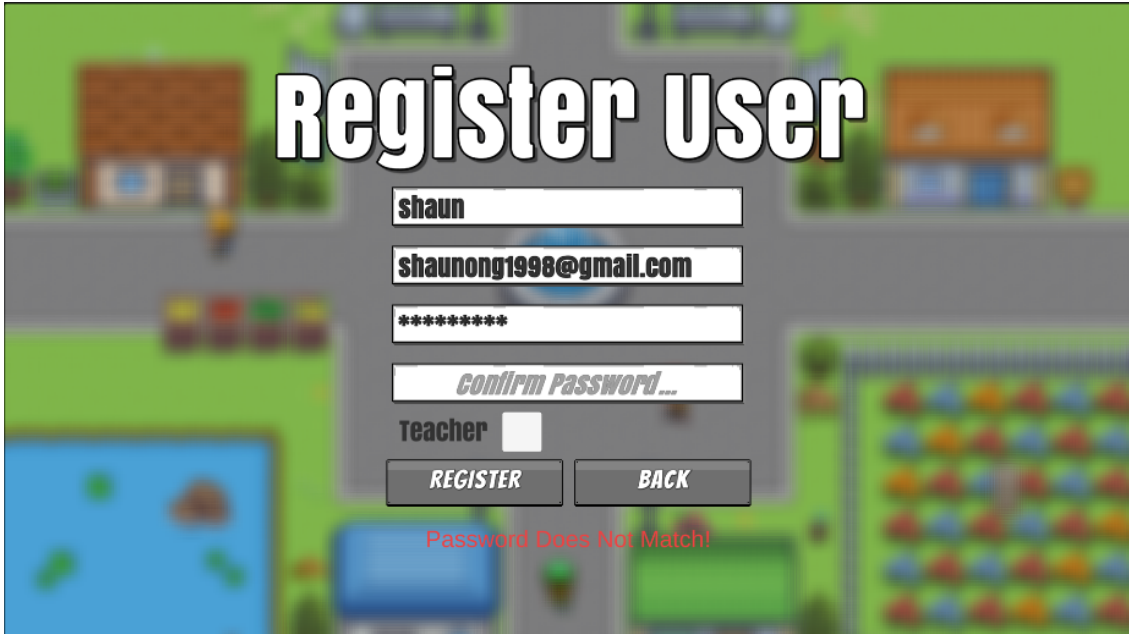
Account Creation Screenshots

Case no.	Output
1	

2	 Register User shaun2 new@gmail.com ***** ***** Teacher ☐ REGISTER BACK Username Taken! Please Choose Another Username!
3	 Register User Enter Username... shaunong1998@gmail.com ***** ***** Teacher ☐ REGISTER BACK Missing Username

4	 The screenshot shows the 'Register User' interface. The title 'Register User' is at the top. Below it are four input fields: the first contains 'Shaun', the second is empty with the placeholder 'Enter Email...', the third contains '*****', and the fourth contains '*****'. There is a 'Teacher' label with an unchecked checkbox. At the bottom are 'REGISTER' and 'BACK' buttons. A red error message 'Missing Email' is displayed below the buttons.
5	 The screenshot shows the 'Register User' interface. The title 'Register User' is at the top. Below it are four input fields: the first contains 'Shaun', the second contains 'testtest', the third contains '*****', and the fourth contains '*****'. There is a 'Teacher' label with an unchecked checkbox. At the bottom are 'REGISTER' and 'BACK' buttons. A red error message 'Register Failed!' is displayed below the buttons.

6	 The screenshot shows a 'Register User' form with a green background. The form contains the following fields: a text input for 'shaun', a text input for 'shaunong1998@gmail.com', two password inputs both showing '*****', and a checkbox labeled 'Teacher'. Below the inputs are two buttons: 'REGISTER' and 'BACK'. A red error message 'Email Already In Use' is displayed at the bottom of the form.
7	 The screenshot shows the same 'Register User' form, but with a different background image. The password inputs now show placeholder text: 'Enter Password...' and 'confirm Password...'. The 'Teacher' checkbox is still present. The 'REGISTER' and 'BACK' buttons are at the bottom. A red error message 'Password Must Contain At Least 6 Characters' is displayed below the buttons.

8	 Register User shaun shaunong1998@gmail.com ***** ***** Teacher ☐ REGISTER BACK Password Must Contain A Lowercase Letter, Uppercase Letter And Number!
9	 Register User shaun shaunong1998@gmail.com ***** confirm Password... Teacher ☐ REGISTER BACK Password Does Not Match!

10	 The screenshot shows the 'Register User' interface with a blurred background of a village scene. The title 'Register User' is at the top. Below it are input fields for 'username' (containing 'shaun'), 'email' (containing 'shaunong1998@gmail.com'), 'password' (containing '*****'), and 'confirm password' (containing '*****'). There is a 'Teacher' checkbox which is unchecked. At the bottom are 'REGISTER' and 'BACK' buttons. A red error message 'Password Does Not Match!' is displayed below the buttons.
11	 The screenshot shows the 'Register User' interface with the same blurred background. The title 'Register User' is at the top. Below it are input fields for 'username' (containing 'teacher'), 'email' (containing 'teacher@gmail.com'), 'password' (containing '*****'), and 'confirm password' (containing '*****'). There is a 'Teacher' checkbox which is checked, and a 'PIN1' input field next to it. At the bottom are 'REGISTER' and 'BACK' buttons. A green success message 'Teacher Account Created Successfully' is displayed below the buttons.

12	
13	

7.2 Login

Equivalence Class

Email

Valid Email	Strings containing character @ and .com
Invalid Email	Strings not containing character @ and .com
Invalid Email	Email does not exist in Firebase
Password	
Valid Password	Matches Password in Firebase
Invalid Password	Does not match Password in Firebase

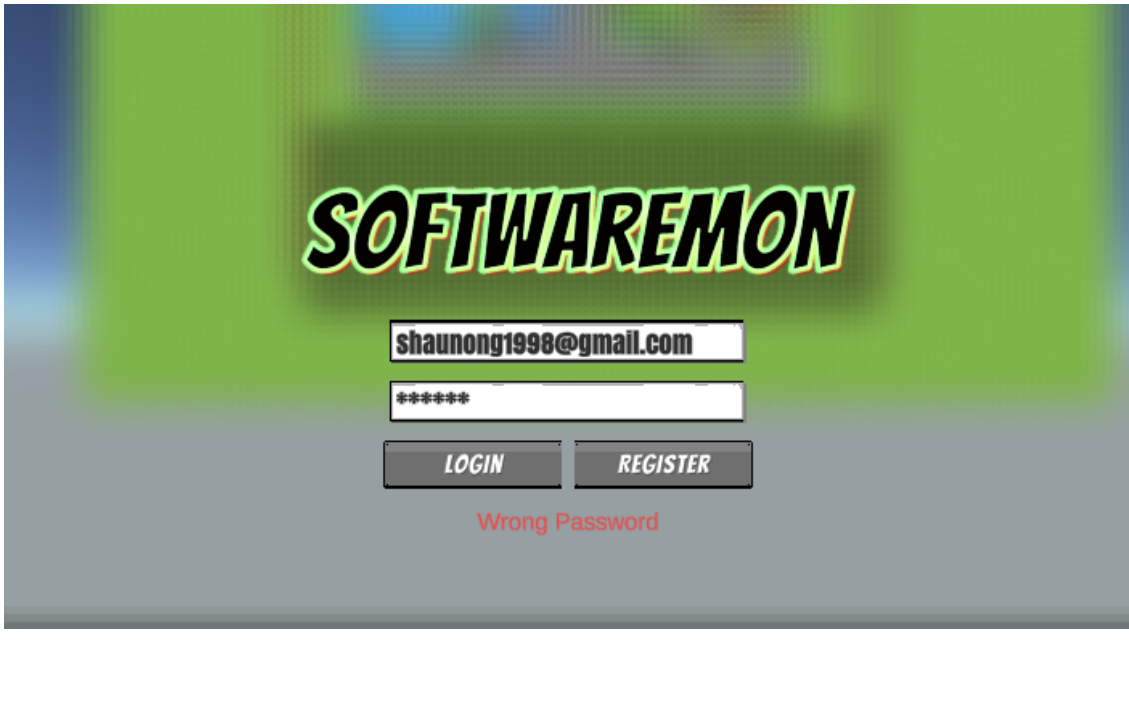
Case no.	Email	Password	Expected Output	Actual Output
1	Valid Email	Password matches Firebase	Login Successful	Login Successful
2	Not Entered	Password matches firebase	Login Unsuccessful	Login Unsuccessful
3	Invalid Email Format	Password matches Firebase	Login Unsuccessful	Login Unsuccessful
4	Email does not exist in firebase	Password matches Firebase	Login Unsuccessful	Login Unsuccessful
5	Valid Email	Not Entered	Login Unsuccessful	Login Unsuccessful
6	Valid Email	Password does not match Firebase	Login Unsuccessful	Login Unsuccessful

Login Screenshots

Case no.	Output
----------	--------

1	 The screenshot shows the SoftwareMon login interface. At the top, the word "SOFTWAREMON" is displayed in a large, bold, black font with a green and yellow outline. Below the title, there are two input fields: the first contains the email address "shaunong1998@gmail.com" and the second contains a masked password "*****". Below these fields are two buttons: "LOGIN" and "REGISTER". Below the buttons, the text "Logged in" is displayed in green.
2	 The screenshot shows the SoftwareMon login interface. At the top, the word "SOFTWAREMON" is displayed in a large, bold, black font with a green and yellow outline. Below the title, there are two input fields: the first contains the placeholder text "Enter Email ..." and the second contains a masked password "*****". Below these fields are two buttons: "LOGIN" and "REGISTER". Below the buttons, the text "Missing Email" is displayed in red.

3	 The screenshot shows the SoftwareMon login interface. The title "SOFTWAREMON" is displayed in a large, stylized, green and yellow font. Below the title are two input fields: the first contains "testtest" and the second contains "*****". Below these fields are two buttons: "LOGIN" and "REGISTER". A red error message "Invalid Email" is displayed below the buttons.
4	 The screenshot shows the SoftwareMon login interface. The title "SOFTWAREMON" is displayed in a large, stylized, green and yellow font. Below the title are two input fields: the first contains "hello@gmail.com" and the second contains "*****". Below these fields are two buttons: "LOGIN" and "REGISTER". A red error message "Account does not exist" is displayed below the buttons.

5	 The screenshot shows the SoftwareMon login interface. At the top, the word "SOFTWAREMON" is displayed in a large, stylized, black font with a green and yellow outline. Below the title, there are two input fields: the first contains the email address "shaunong1998@gmail.com", and the second is labeled "Enter Password...". Below these fields are two buttons: "LOGIN" and "REGISTER". At the bottom, the text "Missing Password" is displayed in red.
6	 The screenshot shows the SoftwareMon login interface. At the top, the word "SOFTWAREMON" is displayed in a large, stylized, black font with a green and yellow outline. Below the title, there are two input fields: the first contains the email address "shaunong1998@gmail.com", and the second contains six asterisks "*****". Below these fields are two buttons: "LOGIN" and "REGISTER". At the bottom, the text "Wrong Password" is displayed in red.

7.3 View Statistics

Equivalence Class

World	
Valid Input	OODP
Valid Input	SE
Valid Input	SSAD
Section	
Valid Input	1
Valid Input	2
Valid Input	3

Case No.	World	Section	Expected Output	Actual Output
1	OODP	1	Display 1.1 stats	Display 1.1 stats
2	OODP	2	Display 1.2 stats	Display 1.2 stats
3	OODP	3	Display 1.3 stats	Display 1.3 stats
4	SE	1	Display 2.1 stats	Display 2.1 stats
5	SE	2	Display 2.2 stats	Display 2.2 stats
6	SE	3	Display 2.3 stats	Display 2.3 stats
7	SSAD	1	Display 3.1 stats	Display 3.1 stats
8	SSAD	2	Display 3.2 stats	Display 3.2 stats
9	SSAD	3	Display 3.3 stats	Display 3.3 stats

Display User Data Screenshots

Case no.	Output
----------	--------

1	
6	

7.4 Leaderboard

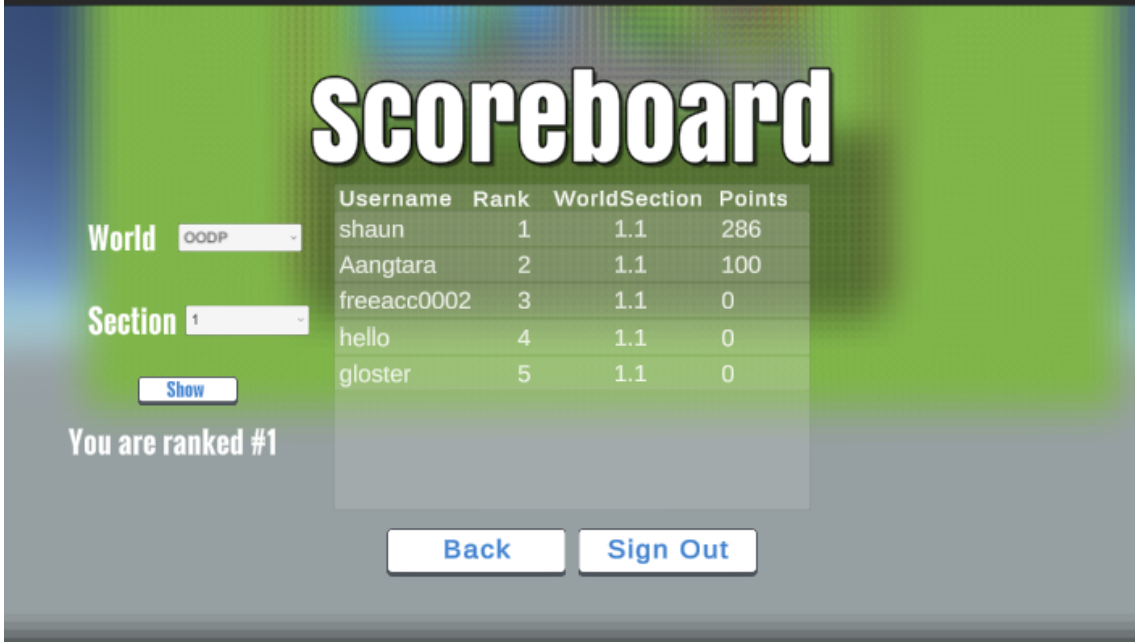
Equivalence Class

World	
Valid Input	OODP
Valid Input	SE
Valid Input	SSAD
Section	
Valid Input	1
Valid Input	2
Valid Input	3

Case No.	World	Section	Expected Output	Actual Output
1	OODP	1	Display 1.1 points	Display 1.1 points
2	OODP	2	Display 1.2 points	Display 1.2 points
3	OODP	3	Display 1.3 points	Display 1.3 points
4	SE	1	Display 2.1 points	Display 2.1 points
5	SE	2	Display 2.2 points	Display 2.2 points
6	SE	3	Display 2.3 points	Display 2.3 points
7	SSAD	1	Display 3.1 points	Display 3.1 points
8	SSAD	2	Display 3.2 points	Display 3.2 points
9	SSAD	3	Display 3.3 points	Display 3.3 points

Leaderboard Screenshots

Case no.	Output
----------	--------

1	
6	

7.5 Create Custom Room

Equivalence Class

RoomID

Valid Input	RoomID does not exist in firebase
Invalid Input	RoomID already exists in firebase
World	
Valid Input	OODP
Valid Input	SE
Valid Input	SSAD
Section	
Valid Input	1
Valid Input	2
Valid Input	3
Difficulty	
Valid Input	Easy
Valid Input	Medium
Valid Input	Hard

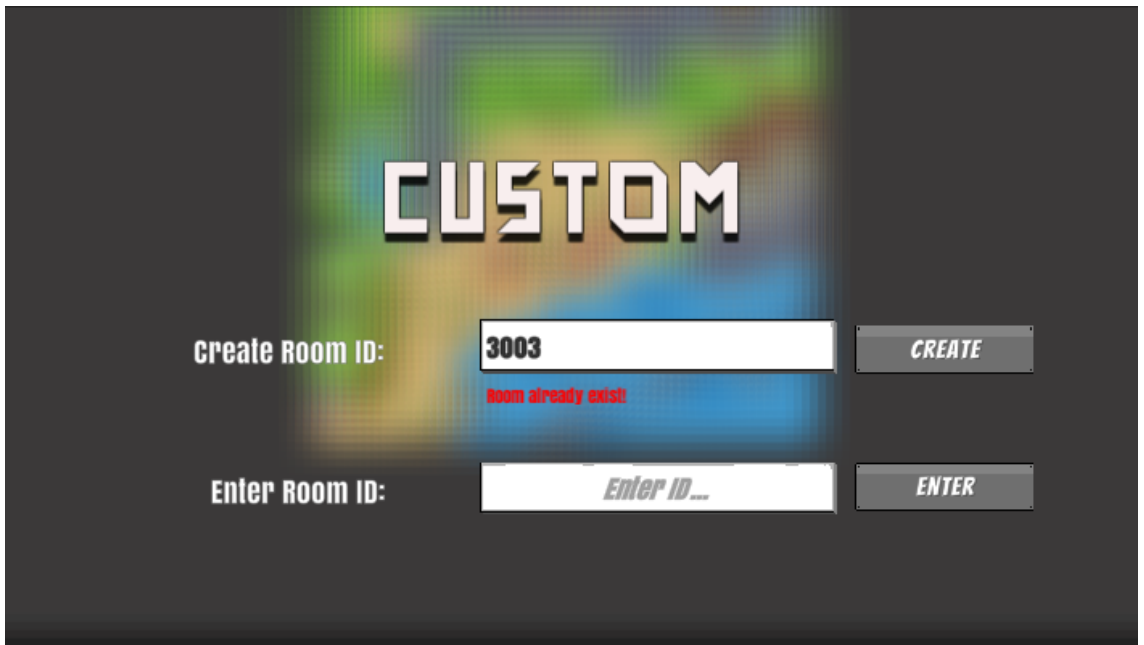
Case No.	RoomID	World	Section	Difficulty	Expected Output	Actual Output
1	RoomID already exists in Firebase	Not Applicable	Not Applicable	Not Applicable	Room Creation Unsuccessful	Room Creation Unsuccessful
2	RoomID does not exist in Firebase	OODP	1	Easy	Display 1.1 Easy questions	Display 1.1 Easy questions
3	RoomID does not exist in Firebase	OODP	1	Medium	Display 1.1 Medium questions	Display 1.1 Medium questions
4	RoomID does not exist in	OODP	1	Hard	Display 1.1 Hard questions	Display 1.1 Hard questions

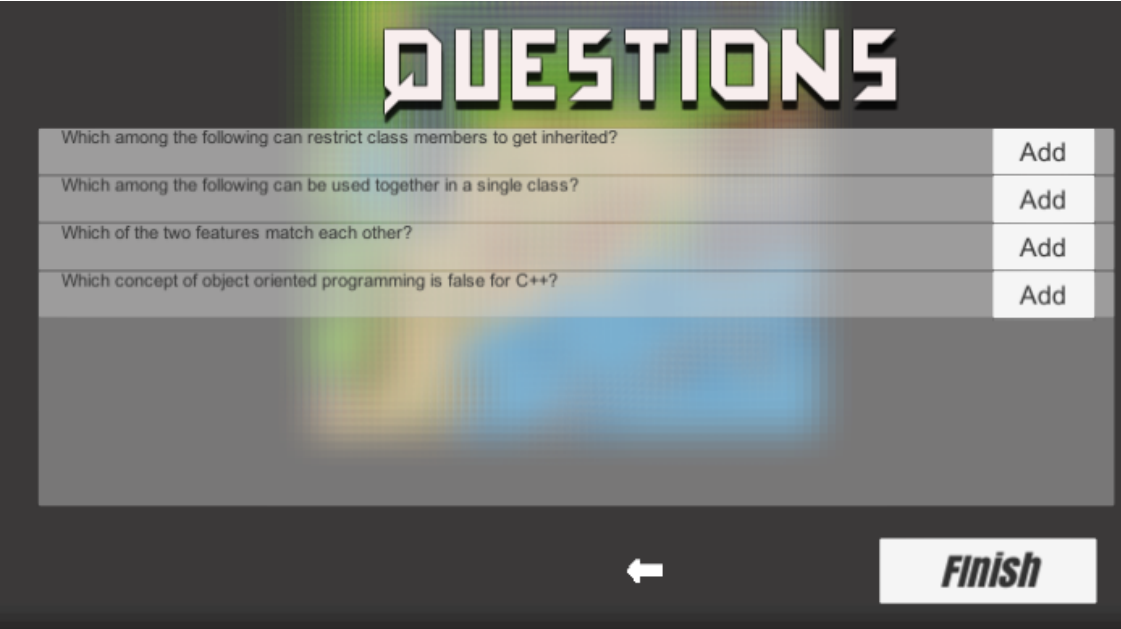
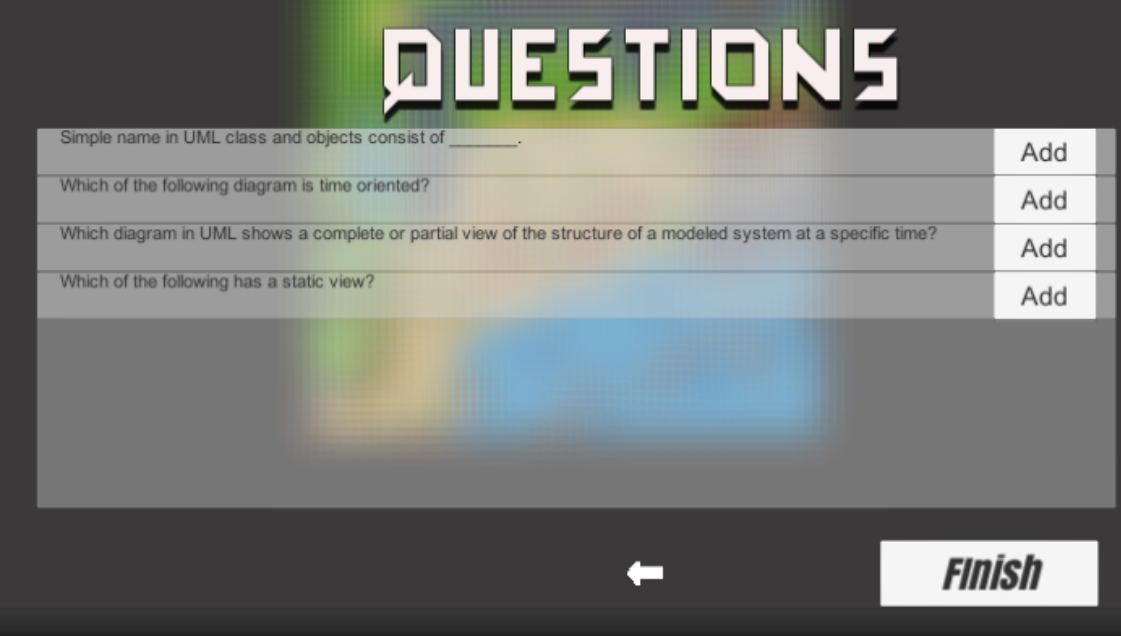
	Firestore					
5	RoomID does not exist in Firestore	OODP	2	Easy	Display 1.2 Easy questions	Display 1.2 Easy questions
6	RoomID does not exist in Firestore	OODP	2	Medium	Display 1.2 Medium questions	Display 1.2 Medium questions
7	RoomID does not exist in Firestore	OODP	2	Hard	Display 1.2 Hard questions	Display 1.2 Hard questions
8	RoomID does not exist in Firestore	OODP	3	Easy	Display 1.3 Easy questions	Display 1.3 Easy questions
9	RoomID does not exist in Firestore	OODP	3	Medium	Display 1.3 Medium questions	Display 1.3 Medium questions
10	RoomID does not exist in Firestore	OODP	3	Hard	Display 1.3 Hard questions	Display 1.3 Hard questions
11	RoomID does not exist in Firestore	SE	1	Easy	Display 2.1 Easy questions	Display 2.1 Easy Questions
12	RoomID does not exist in Firestore	SE	1	Medium	Display 2.1 Medium questions	Display 2.1 Medium questions
13	RoomID does not exist in Firestore	SE	1	Hard	Display 2.1 Hard questions	Display 2.1 Hard questions
14	RoomID does not exist in Firestore	SE	2	Easy	Display 2.2 Easy questions	Display 2.2 Easy questions

	Firestore					
15	RoomID does not exist in Firestore	SE	2	Medium	Display 2.2 Medium questions	Display 2.2 Medium questions
16	RoomID does not exist in Firestore	SE	2	Hard	Display 2.2 Hard questions	Display 2.2 Hard questions
17	RoomID does not exist in Firestore	SE	3	Easy	Display 2.3 Easy questions	Display 2.3 Easy questions
18	RoomID does not exist in Firestore	SE	3	Medium	Display 2.3 Medium questions	Display 2.3 Medium questions
19	RoomID does not exist in Firestore	SE	3	Hard	Display 2.3 Hard questions	Display 2.3 Hard questions
20	RoomID does not exist in Firestore	SSAD	1	Easy	Display 3.1 Easy questions	Display 3.1 Easy questions
21	RoomID does not exist in Firestore	SSAD	1	Medium	Display 3.1 Medium questions	Display 3.1 Medium questions
22	RoomID does not exist in Firestore	SSAD	1	Hard	Display 3.1 Hard questions	Display 3.1 Hard questions
23	RoomID does not exist in Firestore	SSAD	2	Easy	Display 3.2 Easy questions	Display 3.2 Easy questions
24	RoomID does not exist in Firestore	SSAD	2	Medium	Display 3.2 Medium questions	Display 3.2 Medium questions

25	RoomID does not exist in Firebase	SSAD	2	Hard	Display 3.2 Hard questions	Display 3.2 Hard questions
26	RoomID does not exist in Firebase	SSAD	3	Easy	Display 3.3 Easy questions	Display 3.3 Easy questions
27	RoomID does not exist in Firebase	SSAD	3	Medium	Display 3.3 Medium questions	Display 3.3 Medium questions
28	RoomID does not exist in Firebase	SSAD	3	Hard	Display 3.3 Hard questions	Display 3.3 Hard questions

Create Custom Room Screenshots

Case no.	Output
1	

3	
5	

7.6 Enter Custom Room

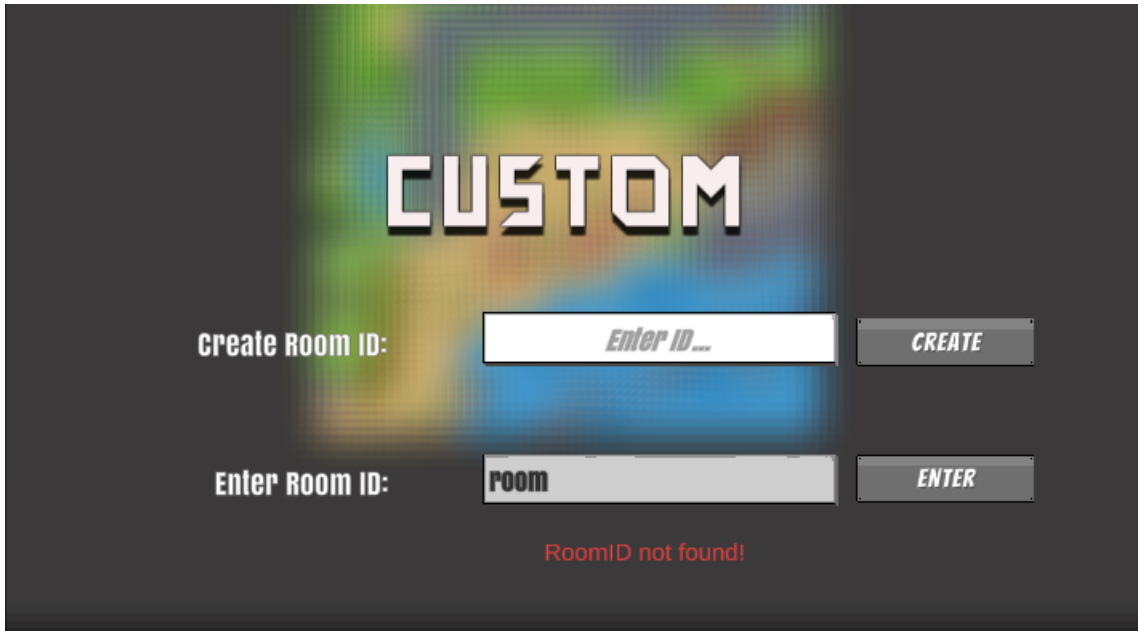
Equivalence Class

RoomID

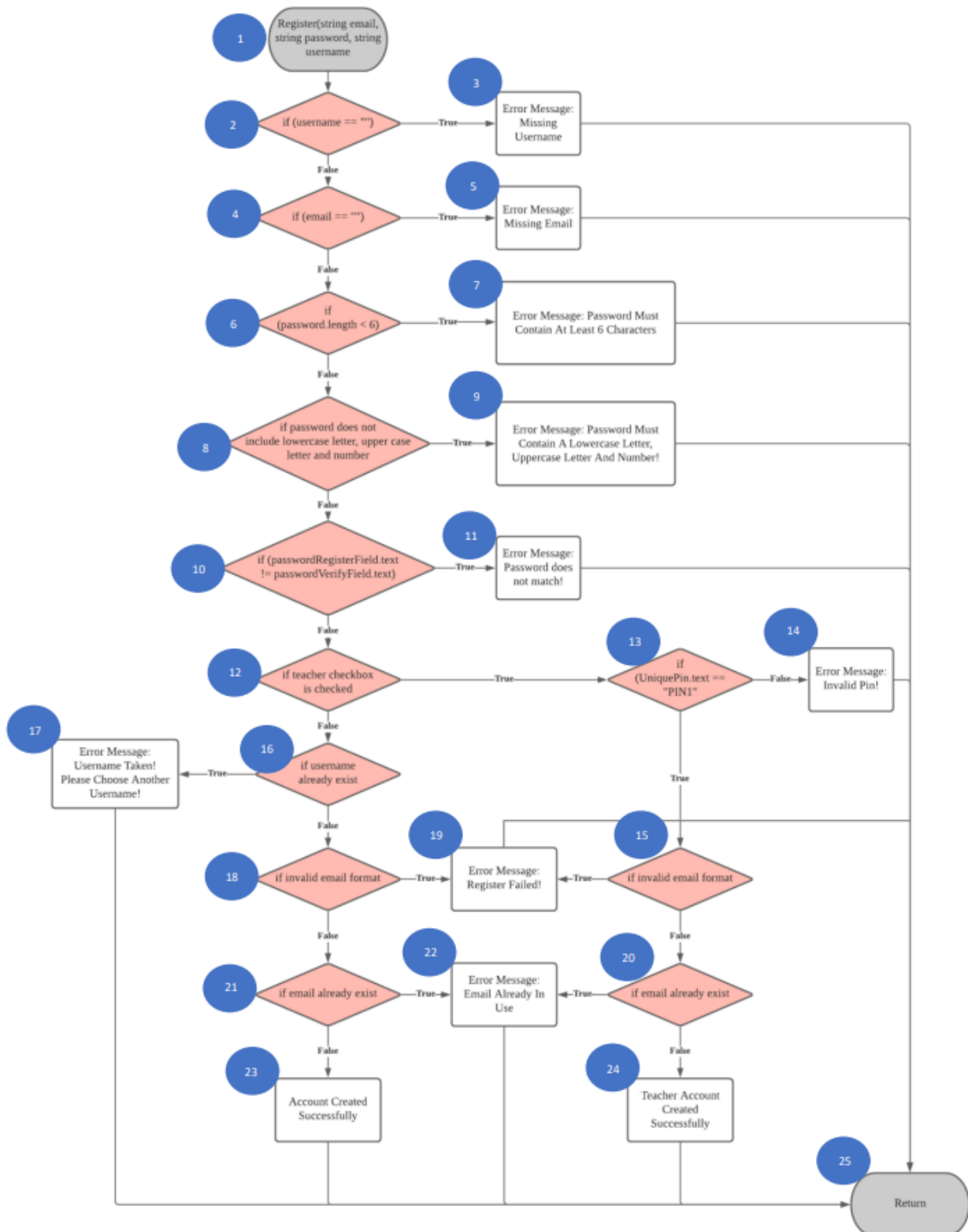
Valid Input	RoomID exists in firebase
Invalid Input	RoomID does not exist in firebase

Case No.	RoomID	Expected Output	Actual Output
1	RoomID exists in Firebase	Room Enter Successful	Room Enter Successful
2	RoomID does not exist in Firebase	Room Enter Unsuccessful	Room Enter Unsuccessful

Enter Custom Room Screenshots

Case no.	Output
2	

8. White-Box Testing

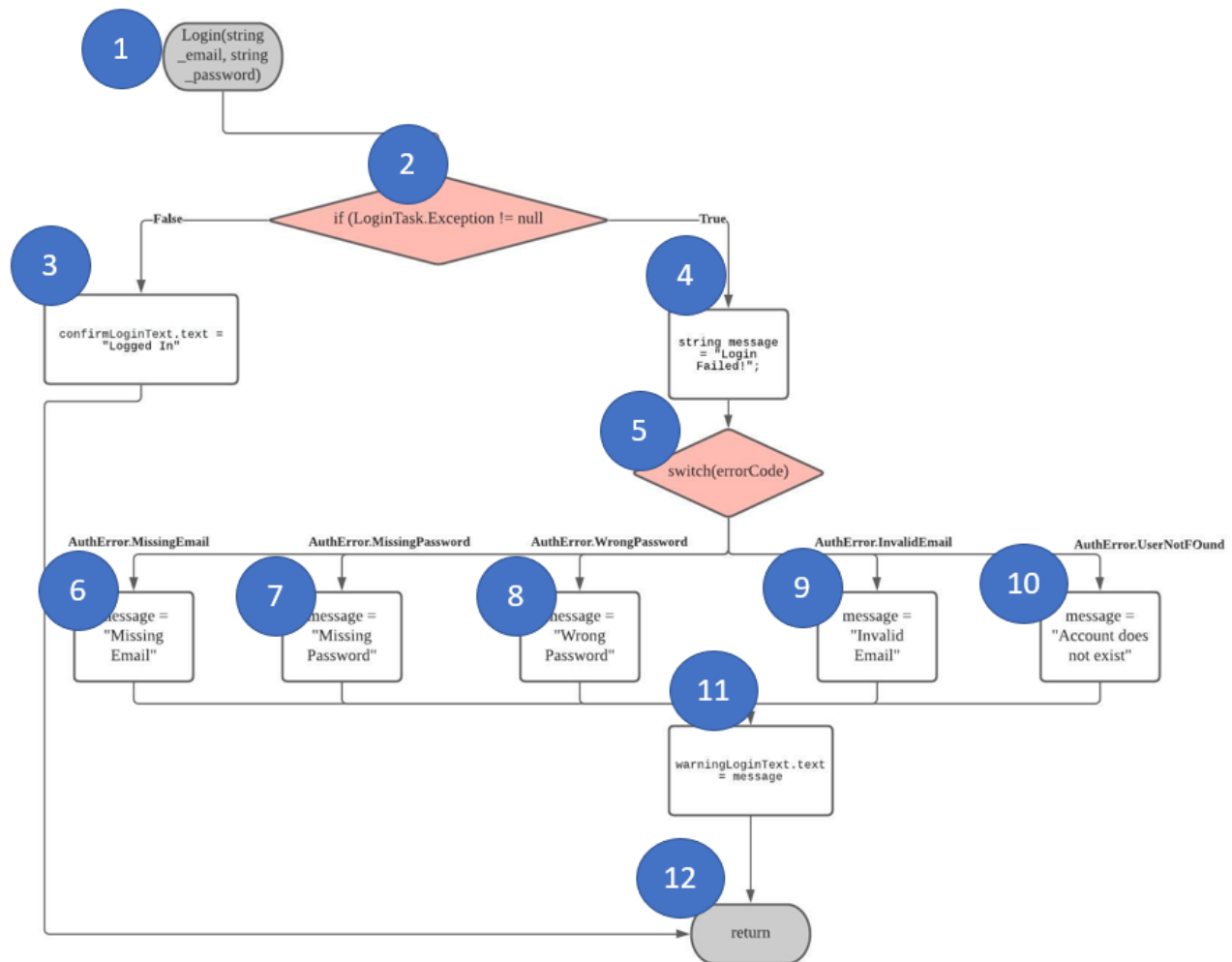


8.1 Account Creation

Cyclomatic Complexity	CC = 36 - 25 + 2 = 13
Basic Path	1,2,4,6,8,10,12,16,18,21,23,25 (Most common path) 1,2,4,6,8,10,12,13,15,20,24,25 1,2,3 1,2,4,5 1,2,4,6,7 1,2,4,6,8,9 1,2,4,6,8,10,11 1,2,4,6,8,10,12,13,14 1,2,4,6,8,10,12,13,15,19 1,2,4,6,8,10,12,13,15,20,22 1,2,4,6,8,10,12,16,17 1,2,4,6,8,10,12,16,18,19 1,2,4,6,8,10,12,16,18,21,22
Test Cases	1. User username is entered, email is entered, password is entered and valid, password match, teacher checkbox is unchecked, username is new, email is valid and new. 2. User username is entered, email is entered, password is entered and valid, password match, teacher checkbox is checked, email is valid and new. 3. User has not entered a username 4. User username is entered, email is missing 5. User username is entered, email is entered, password length is less than 6 6. User username is entered, email is entered, password length is more than 6, password does not contain uppercase and lowercase letter with number 7. User username is entered, email is entered, password length is more than 6, password contain uppercase and lowercase letter with number, password does not match 8. User username is entered, email is entered, password length is more than 6, password contain uppercase and lowercase letter with number, password match, teacher checkbox is checked, unique pin is incorrect

	9. User username is entered, email is entered, password length is more than 6, password contain uppercase and lowercase letter with number, password match, teacher checkbox is checked, unique pin is correct, email format is incorrect 10. User username is entered, email is entered, password length is more than 6, password contain uppercase and lowercase letter with number, password match, teacher checkbox is checked, unique pin is correct, email format is correct, email already exist 11. User username is entered, email is entered, password length is more than 6, password contain uppercase and lowercase letter with number, password match, teacher checkbox is unchecked, username already exist 12. User username is entered, email is entered, password length is more than 6, password contain uppercase and lowercase letter with number, password match, teacher checkbox is unchecked, username is new, email format is incorrect 13. User username is entered, email is entered, password length is more than 6, password contain uppercase and lowercase letter with number, password match, teacher checkbox is unchecked, username is new, email format is correct, email already exist
Real Execution Paths	1. 1,2,4,6,8,10,12,16,18,21,23,25 (Most common path) 2. 1,2,4,6,8,10,12,13,15,20,24,25 3. 1,2,3 4. 1,2,4,5 5. 1,2,4,6,7 6. 1,2,4,6,8,9 7. 1,2,4,6,8,10,11 8. 1,2,4,6,8,10,12,13,14 9. 1,2,4,6,8,10,12,13,15,19 10. 1,2,4,6,8,10,12,13,15,20,22 11. 1,2,4,6,8,10,12,16,17 12. 1,2,4,6,8,10,12,16,18,19 13. 1,2,4,6,8,10,12,16,18,21,22

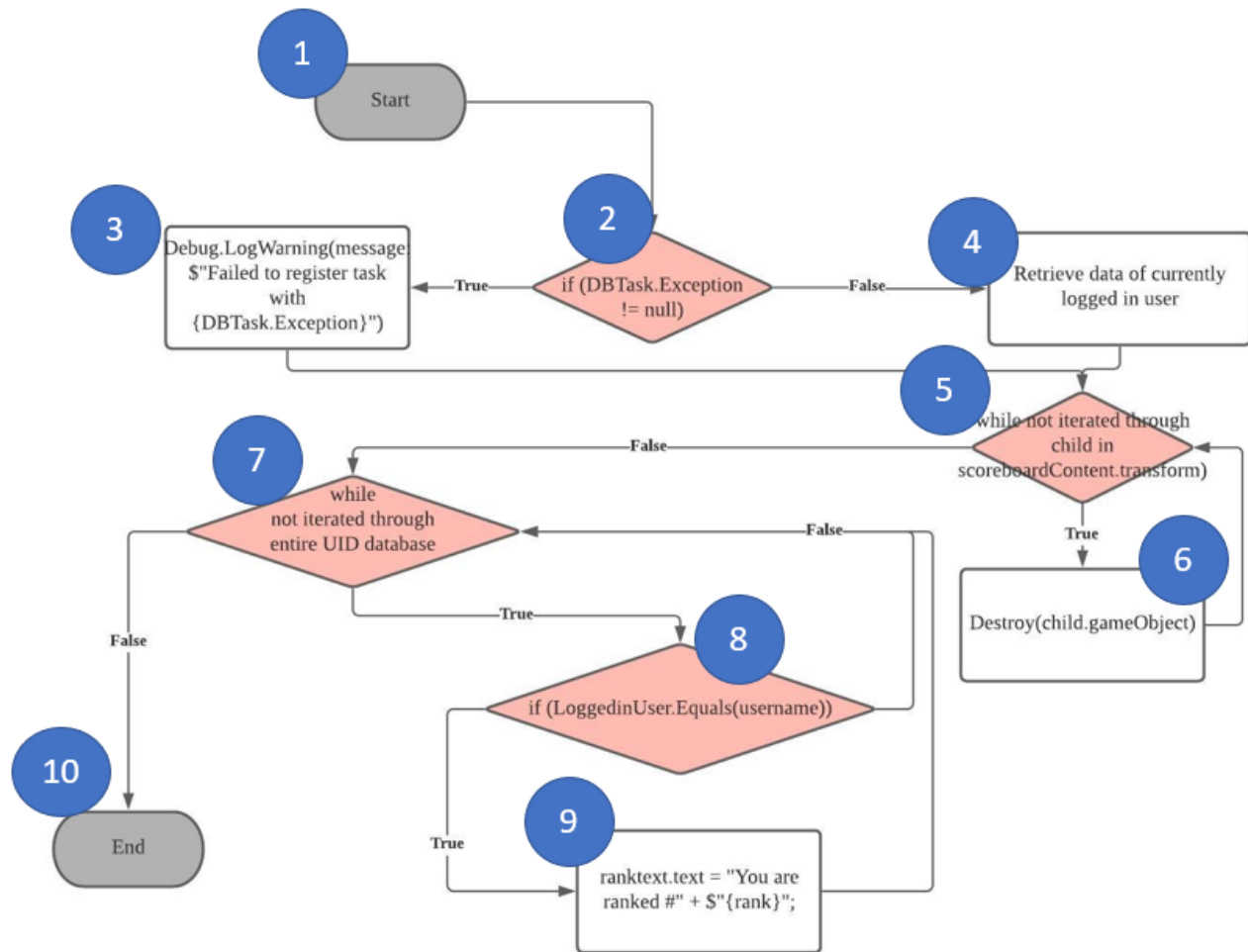
8.2 Login



Cyclomatic Complexity	CC = 16 - 12 + 2 = 6
Basic Path	1. 1,2,3,12 (Most common path) 2. 1,2,4,5,6,11,12 3. 1,2,4,5,7,11,12 4. 1,2,4,5,8,11,12 5. 1,2,4,5,9,11,12 6. 1,2,4,5,10,11,12
Test Cases	1. User's email address and password are accepted and the user is logged into the system.

	2. User login details are not accepted. User has not entered his email. 3. User login details are not accepted. User has not entered his password. 4. User login details are not accepted. User has entered the wrong password. 5. User login details are not accepted. User has entered an invalid email. 6. User login details are not accepted. User account does not exist.
Real Execution Paths	1. 1,2,3,12 (Most common path) 2. 1,2,4,5,6,11,12 3. 1,2,4,5,7,11,12 4. 1,2,4,5,8,11,12 5. 1,2,4,5,9,11,12 6. 1,2,4,5,10,11,12

8.3 Leaderboard



Cyclomatic Complexity	$13 - 10 + 2 = 5$
Basic Path	1. 1,2,4,5,7,10 (Not entering either Loop) 2. 1,2,3,5,7,10 3. 1,2,4,5,6,5,7,10 4. 1,2,4,5,7,8,7,10 5. 1,2,4,5,7,8,9,7,10
Test Cases	1. Data of logged in user is retrieved, no child in scoreboard, UID database is empty 2. Failed to register task, no child in scoreboard, UID database is empty 3. Data of logged in user is retrieved, iterate through scoreboard once and destroy child.gameObject, UID

	database is empty. 4. Data of logged in user is retrieved, no child in scoreboard, cannot find logged in user in UID database 5. Data of logged in user is retrieved, no child in scoreboard, user found in UID database
Real Execution Paths	6. 1,2,4,5,7,10 (Not entering either Loop) 7. 1,2,3,5,7,10 8. 1,2,4,5,6,5,7,10 9. 1,2,4,5,7,8,7,10 10. 1,2,4,5,7,8,9,7,10

9. Integration Testing

9.1 Account Subsystem

9.1.1 Interface Integrity

Case no.	Test Scenario	Components involved	Steps taken	Expected Output	Actual Output
1	Verify that user is able to access Register_UI from Login_UI.	Login_UI, UI_Manager, Register_UI	1.User clicks on the 'Register' button in Login_UI.	User is brought to Register_UI.	User is brought to Register_UI.
2	Verify that user is able to access Login_UI from Register_UI.	Login_UI, UI_Manager, Register_UI	1. User clicks on the 'Back' button in Register_UI.	User is brought to Login_UI.	User is brought to Login_UI.

9.1.2 Functional and End-to-End Validity

Case no.	Test Scenario	Components involved	Steps taken	Expected Output	Actual Output
1	Verify that user is able to register a new user account.	Register_UI, FirebaseManager, FirebaseDatabase	1.User enters registration details. 2.User clicks on the 'Register' button in Register_UI.	FirebaseManager registers user and the new user details are added into the FirebaseDatab	FirebaseManager registers user and the new user details are added into the FirebaseDatab

				ase.	ase.
2	Verify that user is able to login into his user account.	Login_UI, FirebaseManager, FirebaseDatabase	1.User enters login credentials. 2.User clicks on the 'Login' button in Login_UI.	FirebaseManager checks user credentials are valid and proceeds to log in the user	FirebaseManager checks user credentials are valid and proceeds to log in the user.

9.2 Profile Settings Subsystem

9.2.1 Interface Integrity

Case no.	Test Scenario	Components involved	Steps taken	Expected Output	Actual Output
1	Verify that user is able to access UserData_UI from Login_UI.	Login_UI, User_UI	1.User clicks on the 'Login' button in Login_UI.	User is brought to UserData_UI.	User is brought to UserData_UI.

9.2.2 Functional and End-to-End Validity

Case no.	Test Scenario	Components involved	Steps taken	Expected Output	Actual Output
1	Verify that user is able to display his profile data.	UserData_UI, FirebaseManager, FirebaseDatabase	1.User logs in to his user account.	FirebaseManager retrieves data of the currently logged in user from FirebaseDatabase	FirebaseManager retrieves data of the currently logged in user from FirebaseDatabase
2	Verify that user is able to display his profile data according to the world and section.	UserData_UI, FirebaseManager, FirebaseDatabase	1.User logs in to his user account. 2.User chooses the world and section and presses 'show' button in UserData_UI.	FirebaseManager retrieves data of the currently logged in user from FirebaseDatabase according	FirebaseManager retrieves data of the currently logged in user from FirebaseDatabase according

				to specified world and section.	to specified world and section.
--	--	--	--	---------------------------------	---------------------------------

9.3 Leaderboard Subsystem

9.3.1 Interface Integrity

Case no.	Test Scenario	Components involved	Steps taken	Expected Output	Actual Output
1	Verify that user is able to access Scoreboard_UI from UserData_UI	UserData_UI, UI_Manager, Scoreboard_UI	1.User clicks on 'scoreboard' button in UserData_UI.	User is brought to Scoreboard_UI.	User is brought to Scoreboard_UI.
2	Verify that user is able to access UserData_UI from Scoreboard_UI	UserData_UI, UI_Manager, Scoreboard_UI	1.User clicks on 'Back' button in Scoreboard_UI.	User is brought to UserData_UI.	User is brought to UserData_UI.

9.3.2 Functional and End-to-End Validity

Case no.	Test Scenario	Components involved	Steps taken	Expected Output	Actual Output
1	Verify that user is able to view all users' total points.	Scoreboard_UI, FirebaseManager, FirebaseDatabase	1.User logs in to his user account. 2. User goes to view scoreboard.	FirebaseManager retrieves data of all users' total points from FirebaseDatabase and displays them in descending order.	FirebaseManager retrieves data of all users' total points from FirebaseDatabase and displays them in descending order.
2	Verify that user is able to display all users' total points according to world and section.	Scoreboard_UI, FirebaseManager, FirebaseDatabase	1.User logs in to his user account. 2. User goes to view scoreboard. 3.User chooses the world and section and	FirebaseManager retrieves data of all users' total points from FirebaseDatabase	FirebaseManager retrieves data of all users' total points from FirebaseDatabase

			presses 'show' button in Scoreboard_UI.	ase and displays them in descending order according to specified world and section.	ase and displays them in descending order according to specified world and section.
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9.4 Battle Subsystem

9.4.1 Interface Integrity

Case no.	Test Scenario	Components involved	Steps taken	Expected Output	Actual Output
1	Verify that user is able to access BattleScene from MapSelection	MapSelection,World UIManager, BattleScene	1.User clicks on a world. 2. User clicks on a section.	User is brought to BattleScene.	User is brought to BattleScene.
2	Verify that user is able to access MapSelection from BattleScene	MapSelection, WorldUIManager, BattleScene	1.User clicks on 'Run' button in BattleScene.	User is brought to MapSelection.	User is brought to MapSelection.

9.4.2 Functional and End-to-End Validity

Case no.	Test Scenario	Components involved	Steps taken	Expected Output	Actual Output
1	Verify that user is able to update Points in the specified world and section layer as well as TotalPoints.	BattleScene, BattleSystem, FirebaseDatabase	1.User finishes battle with a new high score for that world and section layer.	BattleSystem updates Points in the specified world and section layer as well as TotalPoints in FirebaseDatabase.	BattleSystem updates Points in the specified world and section layer as well as TotalPoints in FirebaseDatabase.

10. System Testing

Preconditions:

- Integration testing
- All units should be integrated as one entire system.
- Device should be in working condition.

Dependencies:

- Entire system

Test Scenario 1

(Assume new user)

1. User registers a new user account.
2. User logs into his new user account.
3. User enters map selection menu.
4. User chooses to play the quest mode.
5. User chooses world OODP and section 1 for his quest mode.
6. User moves and fights a trainer.
7. User defeats the trainer.
8. User pauses the game and exits quest mode.
9. User exits map selection menu and views his profile data.
10. User selects world OODP and section 1 and presses 'show'.
11. User clicks on leaderboard.
12. User clicks on sign out.

Test Scenario 2

(Assume existing user)

1. User logs into his existing user account.
2. User views his profile data.
3. User enters map selection menu.
4. User chooses to play the PVP mode.
5. User chooses world SE and section 3 for his challenge.
6. User defeats the challenge.
7. User exits map selection menu and views his profile data.
8. User selects world SE and section 3 and presses 'show'.
9. User clicks on leaderboard.
10. User selects world SE and section 3 and presses 'show'.
11. User clicks on sign out.

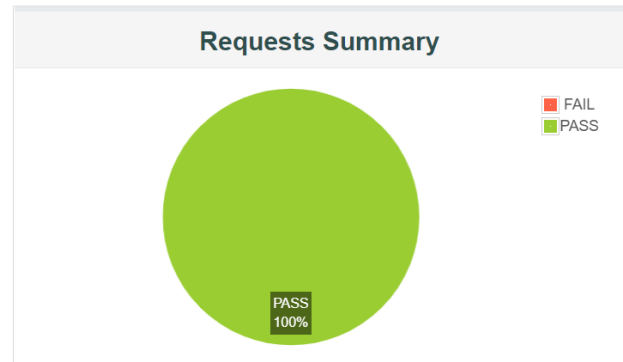
11. Load Testing

Aim: To ensure that the game performs well under their expected workload.

Verify: Check that app does not crash or fail to perform its tasks.

Tools: Apache JMeter

APDEX (Application Performance Index)			
Apdex	T (Toleration threshold)	F (Frustration threshold)	Label
Apdex	T (Toleration threshold)	F (Frustration threshold)	Label
0.784	500 ms	1 sec 500 ms	Total
0.350	500 ms	1 sec 500 ms	Quora
0.415	500 ms	1 sec 500 ms	Quora -2
0.500	500 ms	1 sec 500 ms	Twitter
0.580	500 ms	1 sec 500 ms	Facebook
0.870	500 ms	1 sec 500 ms	Quora -1
0.925	500 ms	1 sec 500 ms	Twitter-2
0.985	500 ms	1 sec 500 ms	Facebook-1
1.000	500 ms	1 sec 500 ms	Twitter-0
1.000	500 ms	1 sec 500 ms	Twitter-1
1.000	500 ms	1 sec 500 ms	Quora -0
1.000	500 ms	1 sec 500 ms	Facebook-0



Statistics

Requests	Executions			Response Times (ms)							Throughput	Network (KB/sec)	
Label	#Samples	FAIL	Error %	Average	Min	Max	Median	90th pct	95th pct	99th pct	Transactions/s	Received	Sent
Requests	Executions			Response Times (ms)							Throughput	Network (KB/sec)	
Label	#Samples	FAIL	Error %	Average	Min	Max	Median	90th pct	95th pct	99th pct	Transactions/s	Received	Sent
Total	1100	0	0.00%	517.77	9	3968	293.00	1081.80	1426.00	3089.51	62.81	4691.73	11.11
Facebook	100	0	0.00%	552.02	463	887	523.00	640.60	691.00	886.18	7.24	1361.56	1.67
Facebook-0	100	0	0.00%	210.83	167	312	205.00	232.90	238.95	311.47	7.41	2.58	0.85
Facebook-1	100	0	0.00%	340.56	280	670	316.50	412.90	479.40	669.34	7.36	1381.45	0.85
Quora	100	0	0.00%	1622.30	830	3968	1291.50	3091.80	3409.00	3964.20	6.30	984.92	2.32
Quora -0	100	0	0.00%	21.02	9	105	13.00	35.70	105.00	105.00	6.84	2.19	0.85
Quora -1	100	0	0.00%	506.91	248	2135	274.00	1346.80	1433.80	2129.94	6.78	4.18	0.84
Quora -2	100	0	0.00%	1092.64	548	2264	1004.50	1806.20	2084.40	2263.05	6.98	1084.35	0.84
Twitter	100	0	0.00%	675.17	568	1062	600.00	895.90	918.75	1061.86	7.04	468.51	2.63
Twitter-0	100	0	0.00%	183.00	171	242	176.00	202.80	214.90	241.99	7.47	4.43	0.94